Theory of Computing:

9. Pushdown Automata

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Outline :

- Revision: Context-Free Languages
- Pushdown Automata
- Constructing PDA
- Notations and Formalism
- Examples
- Converting CFG <---> PDA

Context-Free Languages

- Context-free languages are those that can be **generated** by context-free grammar.
- Example : Language L = { 0^n1^n | n >= 0 }

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  - S → aSa | bSb | **ε**

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- Example : Language L = { 0^n1^n | n >= 0 }
  - S → 0S1 | e

Context-Free Languages

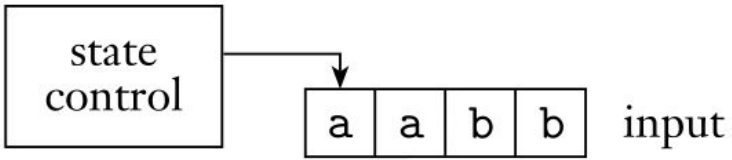
- As with regular languages, there are various approaches to describe the language ：
  - Build Automata that accepts or **recognize**s a string from the language
  - Design Regular expressions that **describe** the language
  - Write the regular grammar to **generate** the language

Context-Free Languages

- For context-free languages :
  - Context-free grammar : to **generate** language.
  - Can we design an automaton to **recognize** a word from the language ?

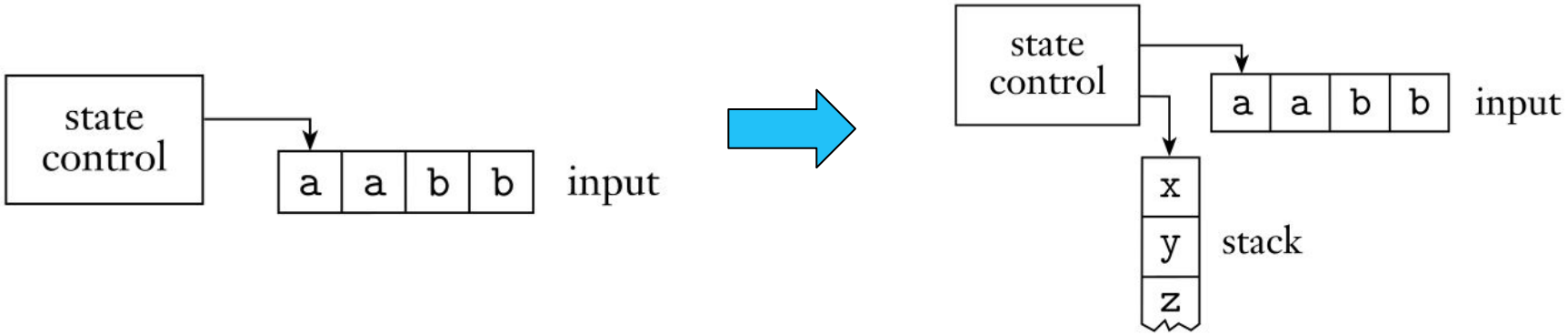
Pushdown Automata

- Can NFA represent a context-free language : ?
- **No :** We need a machine with some extra memory



Pushdown Automata

- Can NFA represent a context-free language : ?
- **No :** We need a machine with some extra memory : **Stack**



Pushdown Automata

- Pushdown Automata (PDA) : can be considered as NFA with a **stack**
- The stack stores information on the **last-in first-out** principle.
  - Items are added on top by **pushing**;
  - items are removed from the top by **popping**
- **Only the top** of the stack is visible at any point in time.

Pushdown Automata

- A pushdown automaton (PDA) has a fixed set of states (like FA), but it also has **one** stack with (theoretically) **infinite** storage.
- When symbol is read, depending on :
  1. *State of automaton*
  2. *Symbol on top of stack*
  3. *Symbol read,*

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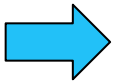

The automaton would:

- *Updates its state*
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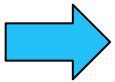


The automaton would:

- *Updates its state*
- *(optionally) **pop**s or **push**es a symbol.*

The automaton may also pop or push without reading input.

Constructing Pushdown Automaton

- **Important notes:**
  - The stack is recommended to be initialized with a special symbol or marker either the **$** or $\mathbf{Z_0}$ symbols
  - The special symbol is used to indicate the bottom of the stack.
  - There are different notations and definitions used for PDA depending on the textbook being used :
    - Sipser's Book :
      - does not have a start stack symbol
      - does not allow transitions to push multiple symbols onto the stack.

Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }

Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = $\{a^n b^n\}$
- **Informal English Description :**

  *Read symbols from the input. As each **a** is read, push it onto the stack.*

  *As soon as **b**s are seen, pop an **a** off the stack for each **b** read.*

  *If reading the input is finished exactly when the stack becomes empty of **a**s, accept the input.*

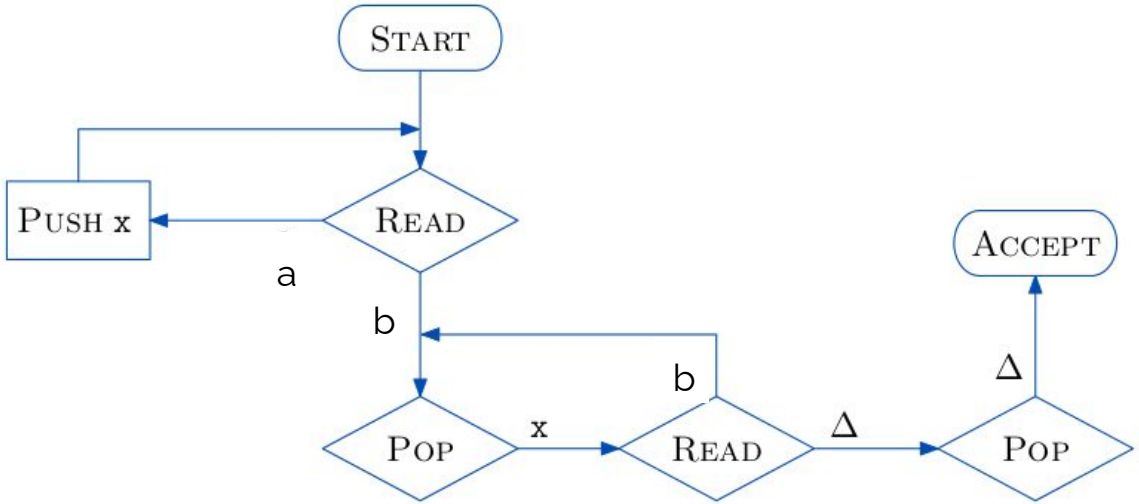
  *If the stack becomes empty while **b**s remain or if the **b**s are finished while the stack still contains **a**s or if any **a**s appear in the input following **b**s, **reject the input***

Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = $\{a^n b^n\}$
- **Design the informal algorithm**
  1. Initialize the Stack with a special marker
  2. while next input character is **a** do
     - push **a**
  3. while next input character is **b** do
     - pop **a**
  4. The special marker is on top of the stack.

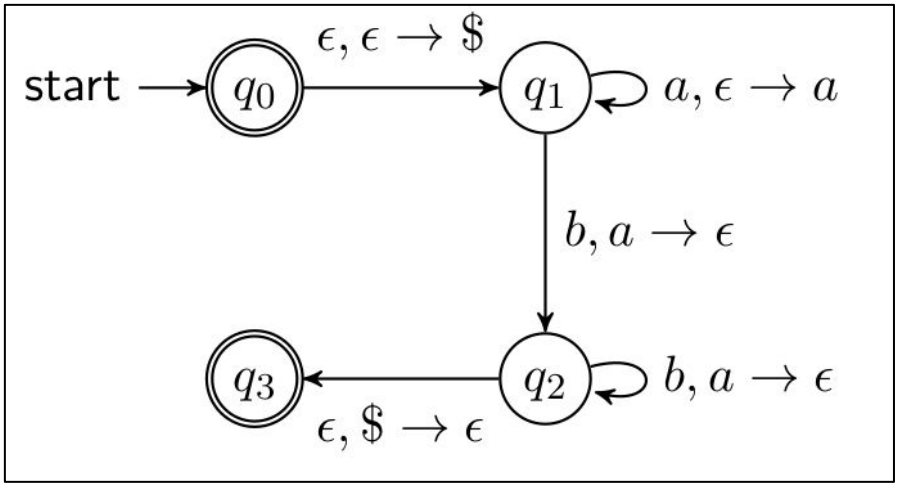
Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = $\{a^n b^n\}$
- **Flowchart** can be used to simplify the concept



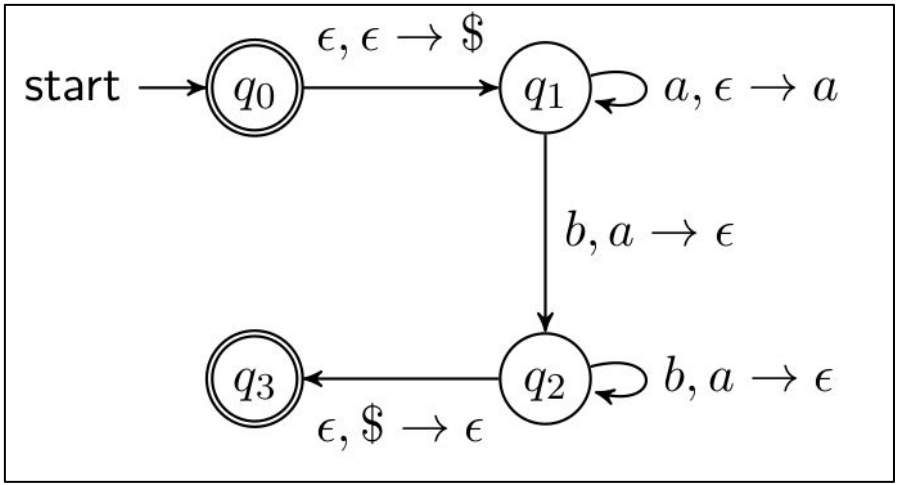
Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {$a^n b^n$ }
- State Diagram



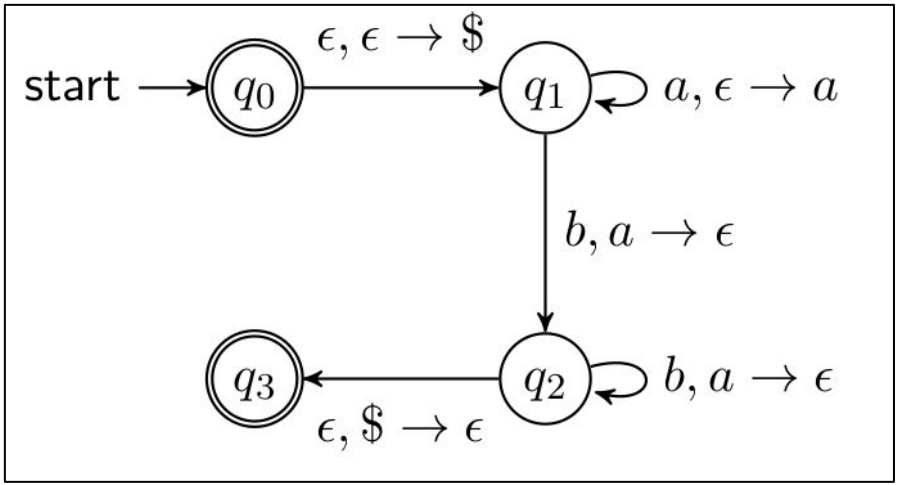
Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- State Diagram
  - **States : q0, q1, q2 , q3**



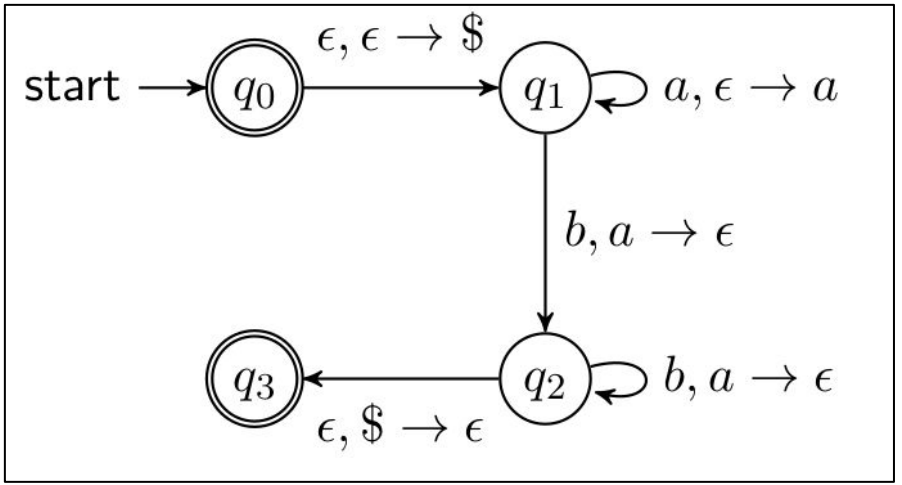
Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- State Diagram
  - **Start State : q0**



Constructing Pushdown Automaton

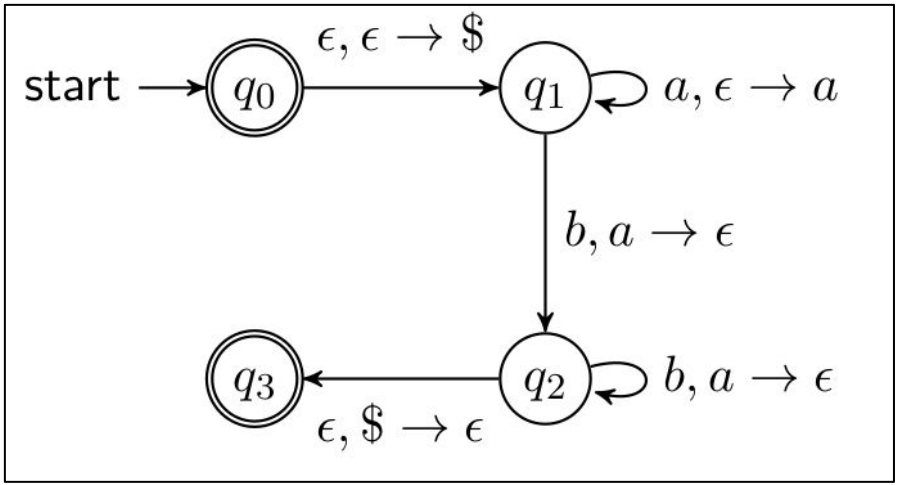
- Let's Construct a PDA that accepts all strings from the language L = {$a^n b^n$ }
- State Diagram
  - **Accepting States : q0 , q3**



Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- State Diagram
  - **Transitions :**

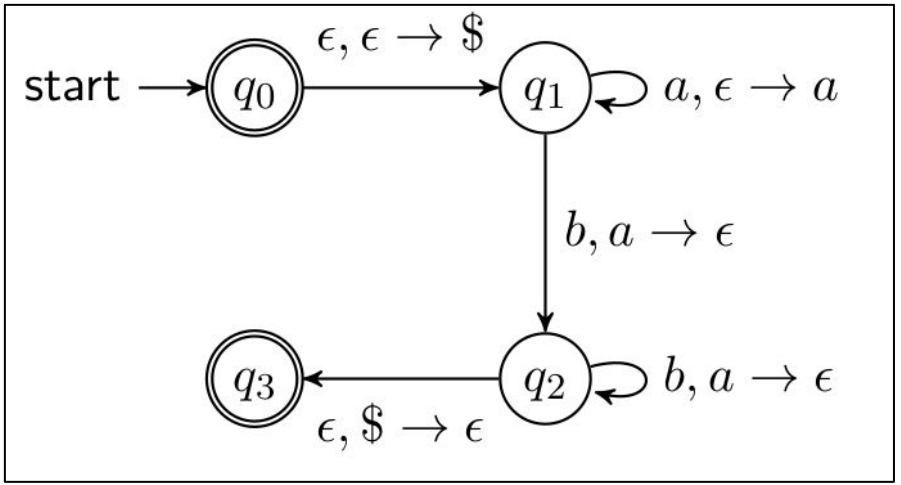
    A , B → C
  - *Means that when*
    - *You read symbol **A**,*
    - *Pop **B***
    - *Push **C***



Constructing Pushdown Automaton

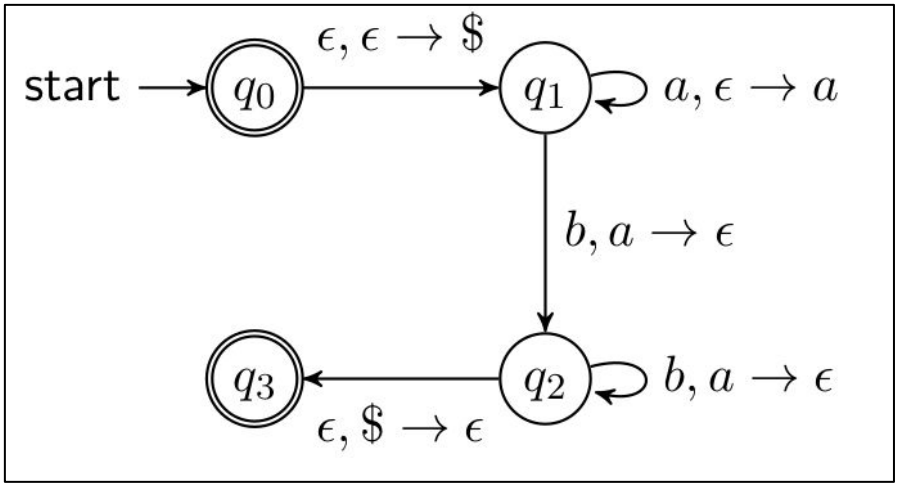
- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- State Diagram
  - **Transitions :**

    A , B → C
  - *Special Cases*
    - **A , ε → B**
      - ***Push B***
    - **A , B → ε**
      - ***Pop B***



Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- State Diagram
  - **First Transition ε, ε → $:**
    - When nothing, place **$** into The top the of stack.
    - **$** is used to serve as a special marker



Constructing Pushdown Automaton

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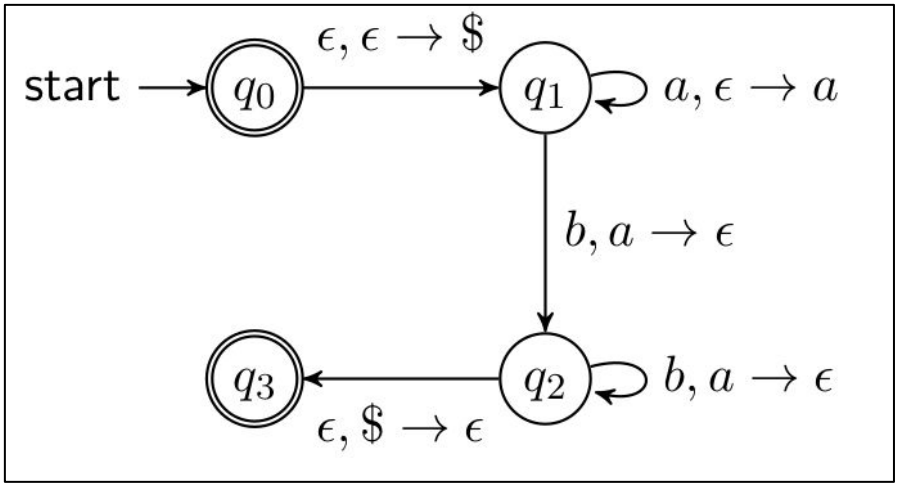
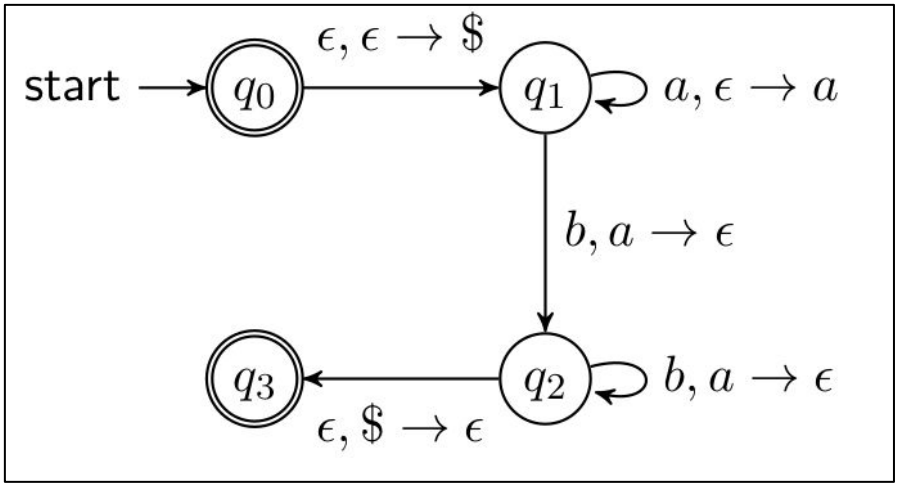
- State Diagram
  - **First Transition ε, ε → $:**
    - (q0, ε, ε) → (q1, $)



Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- Simulation : **aaabbb**



Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
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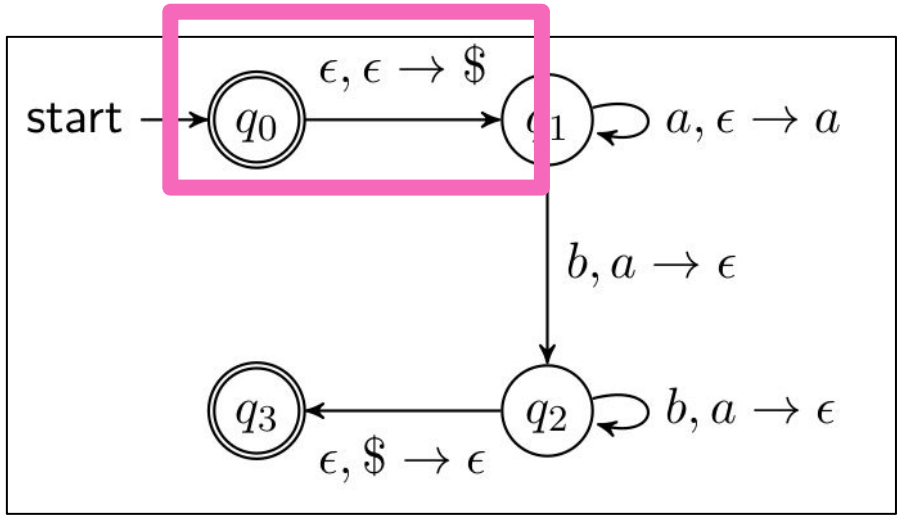
Step	State	Stack	Input	Action
1	q_0		$aaabbb$	push \$



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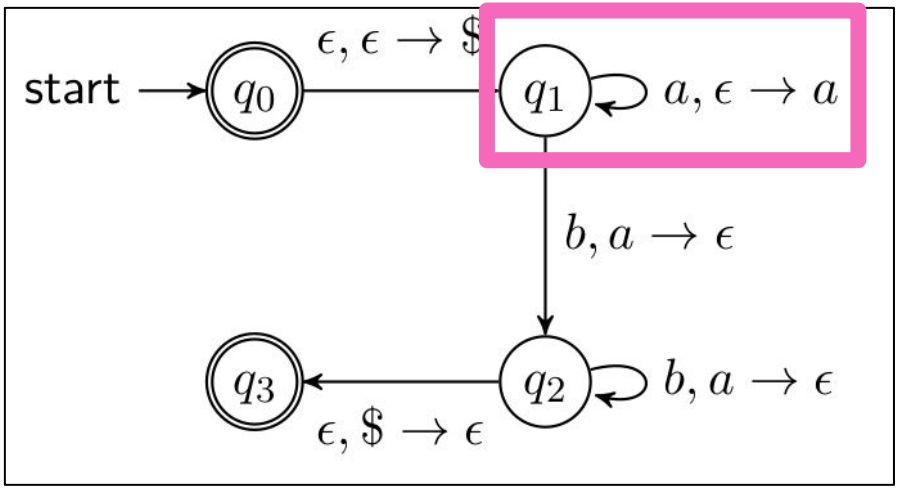

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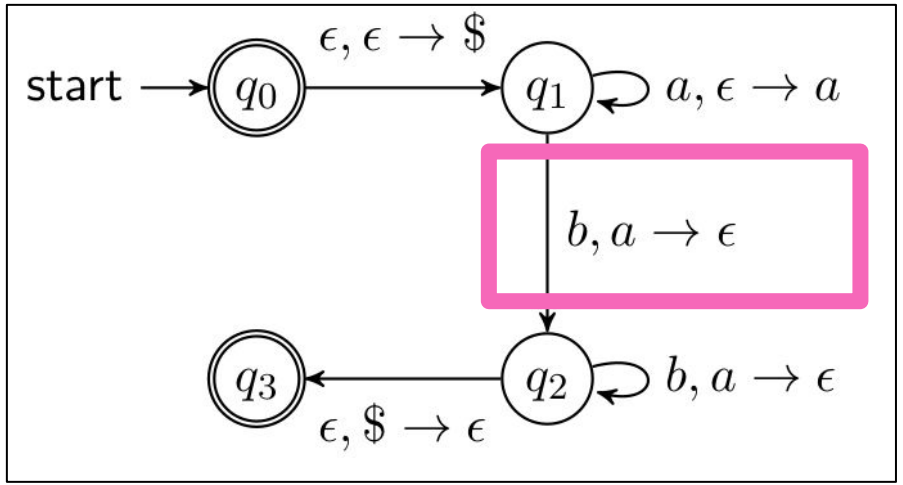
1	q_0		$aaabbb$	push \$
2	q_1	\$	$aaabbb$	push a
3	q_1	\$$a$	$aabbb$	push a
4	q_1	\$$aa$	$abbb$	push a
5	q_1	\$$aaa$	bbb	pop a



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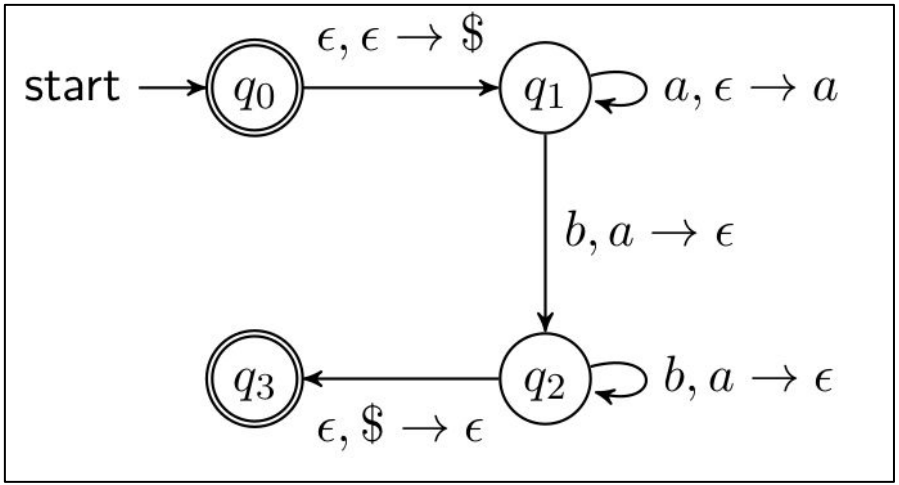
5	q_1	\$$aaa$	bbb	pop a
6	q_2	\$$aa$	bb	pop a
7	q_2	\$$a$	b	pop a
8	q_2	\$		pop \$
9	q_3			accept



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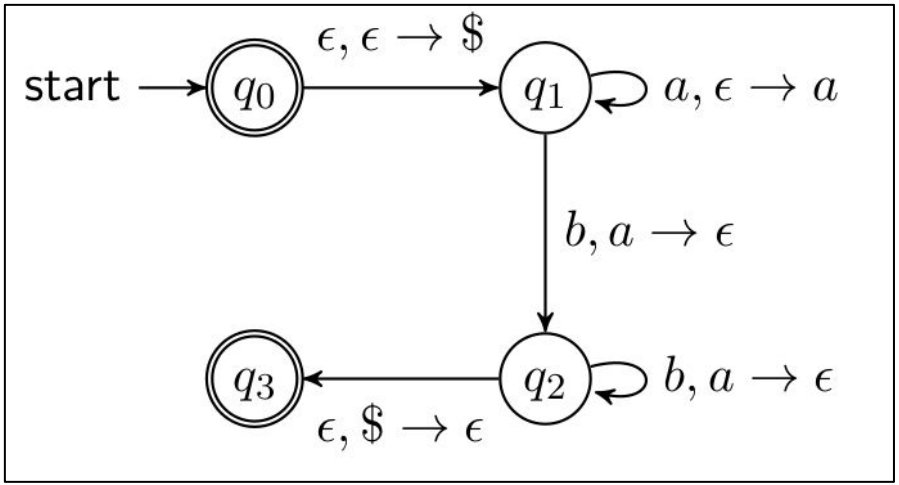
Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- When to accept or reject ?

Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language L = {a^n b^n }
- Simulation : **aababb**

Step	State	Stack	Input	Action
1	q_0		$aababb$	push \$
2	q_1	\$	$aababb$	push a
3	q_1	\$$a$	$ababb$	push a
4	q_1	\$$aa$	$babb$	pop a
5	q_2	\$$a$	abb	crash
6	q_ϕ	\$$a$	bb	
7	q_ϕ	\$$a$	b	
8	q_ϕ	\$$a$		reject



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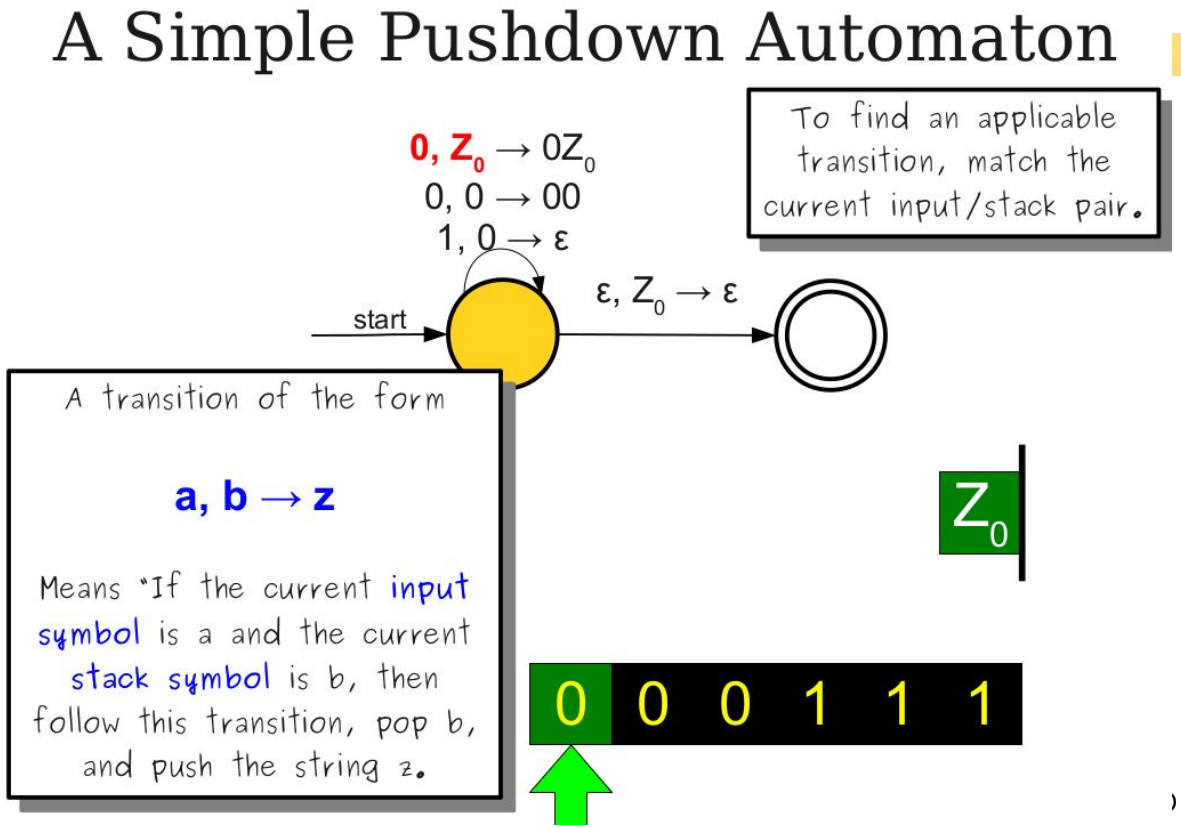
Constructing Pushdown Automaton

- Let's Construct a PDA tha
- Lecture notes from the University of Stanford

Different Notations...



Constructing Pushdown Automaton

- Let's Construct a PDA that accepts all strings from the language $L = \{a^n b^n\}$
- Lecture notes from UNC

$M = (\{q_0, q_1, q_2\}, \{0, 1\}, \{Z_0, Z_1\}, \delta, q_0, Z_0, \{q_2\})$

$\delta(q_0, 0, Z_0) = \{(q_0, Z_1 Z_0)\}$

$\delta(q_0, 0, Z_1) = \{(q_0, Z_1 Z_1)\}$

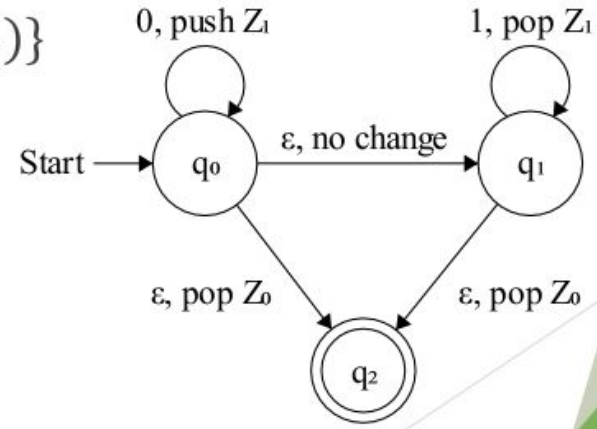
$\delta(q_0, \varepsilon, Z_0) = \{(q_2, \varepsilon), (q_1, Z_0)\}$

$\delta(q_0, \varepsilon, Z_1) = \{(q_1, Z_1)\}$

$\delta(q_1, 1, Z_1) = \{(q_1, \varepsilon)\}$

$\delta(q_1, \varepsilon, Z_0) = \{(q_2, \varepsilon)\}$

Different Notations...

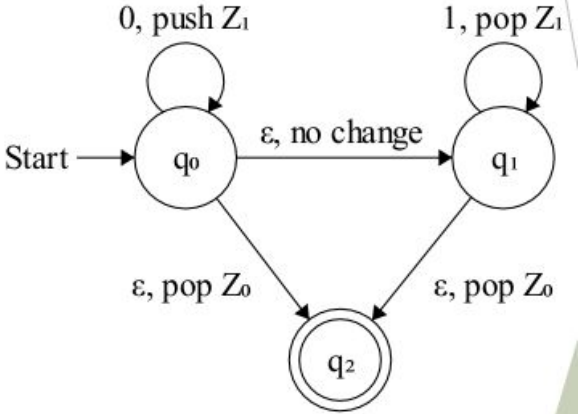


Constructing Pushdown Automaton

Different Notations…

- Let's Construct a PDA that accepts all strings from the language L = $\{a^n b^n\}$
- Lecture notes from UNC
- The symbol : $\vdash$ means : **Goes to**

$$
\begin{aligned}
(q_0, 0011, Z_0) &\vdash (q_0, 011, Z_1 Z_0) \\
&\vdash (q_0, 11, Z_1 Z_1 Z_0) \\
&\vdash (q_1, 11, Z_1 Z_1 Z_0) \\
&\vdash (q_1, 1, Z_1 Z_0) \\
&\vdash (q_1, \varepsilon, Z_0) \\
&\vdash (q_2, \varepsilon, \varepsilon)
\end{aligned}
$$



Constructing Pushdown Automaton

We will use strictly the notation :

a , b→ c

Or

a;b;c

When **reading** a, **pop** b and **push** c

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- $L = \{ww^R \mid w \in \{0,1\}^* \}$

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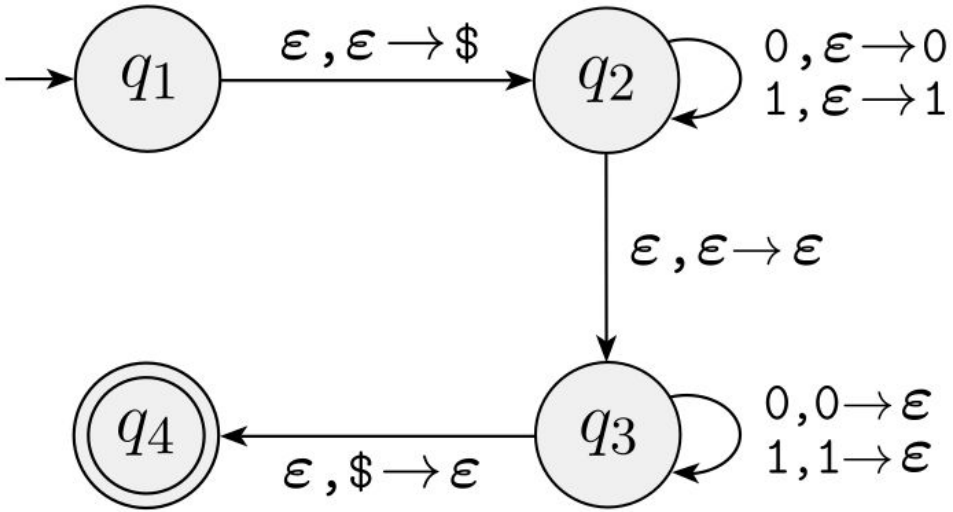
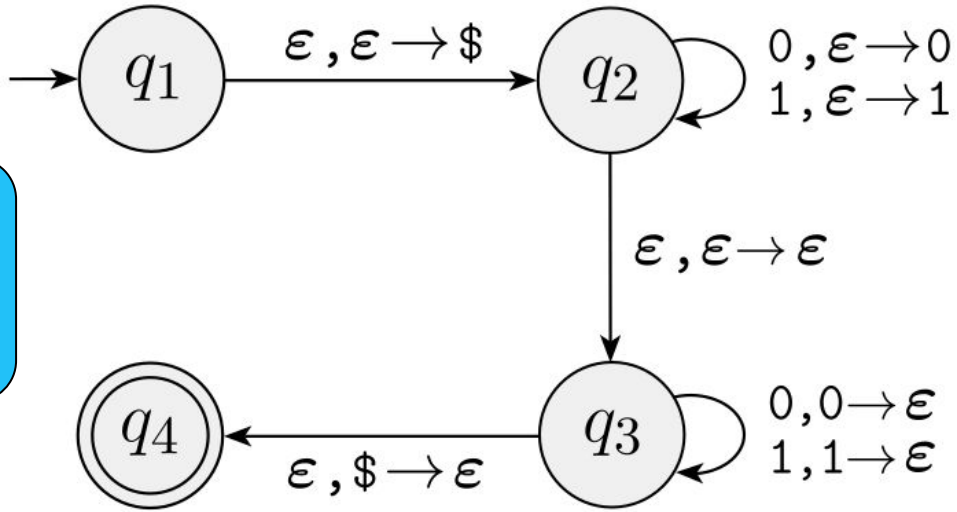


Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- $L = \{ww^R \mid w \in \{0,1\}^*\}$

Let's simulate recognizing the word :
1001



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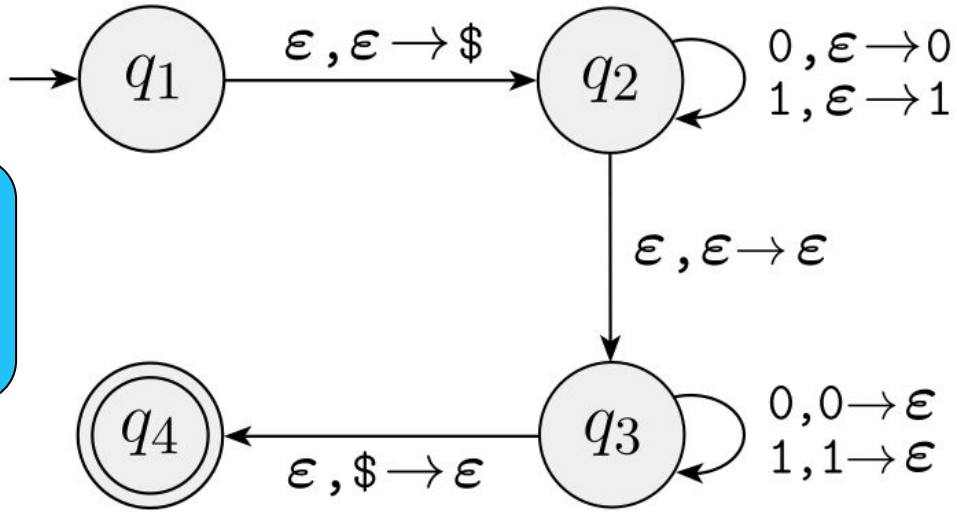
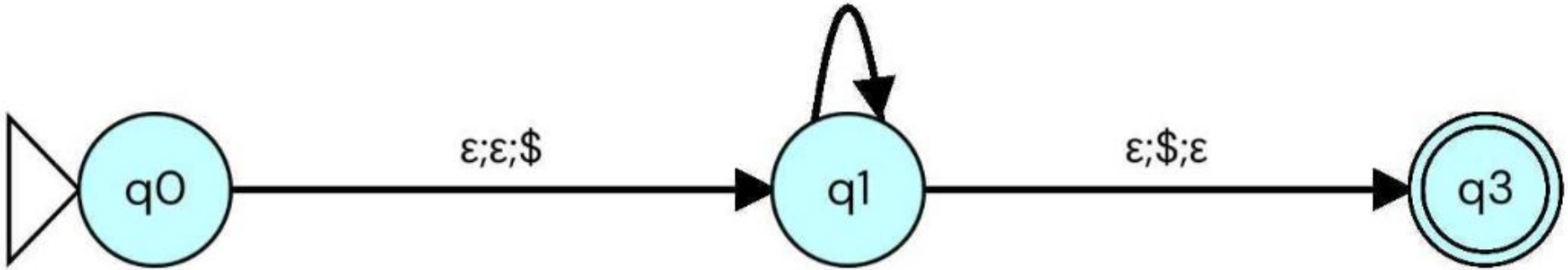


Let's simulate recognizing the word :
0011

Examples for Creating Pushdown Automata

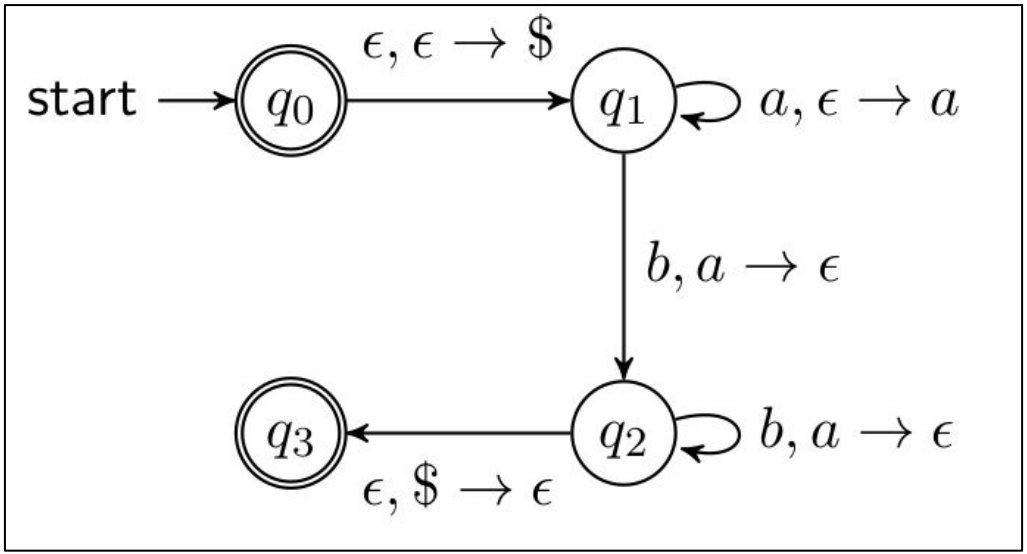
- **Most** PDAs are in the form :

Transition Label 1
Transition Label 2
..



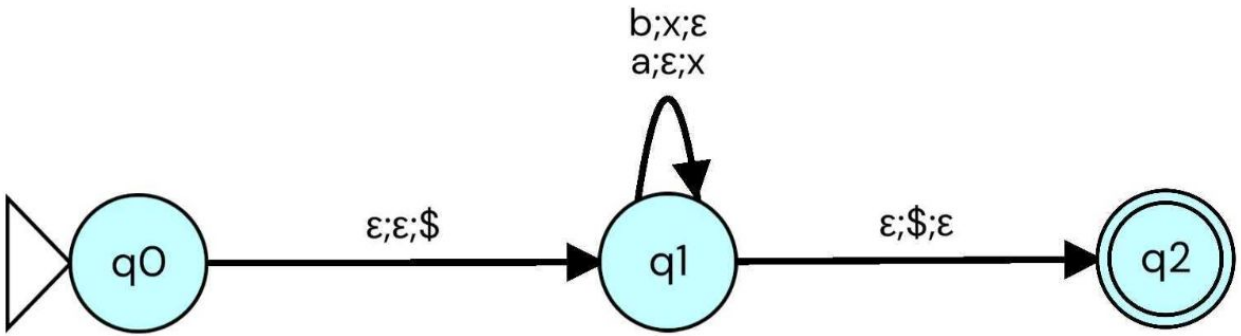
Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- L = {$a^n b^n$ }



Examples for Creating Pushdown Automata

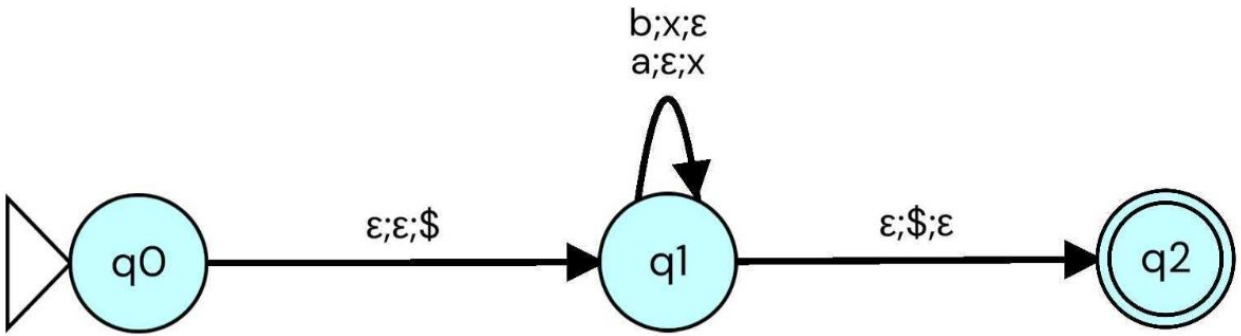
- Let's Construct a PDA that accepts all strings from the language
- L = $\{a^n b^n\}$



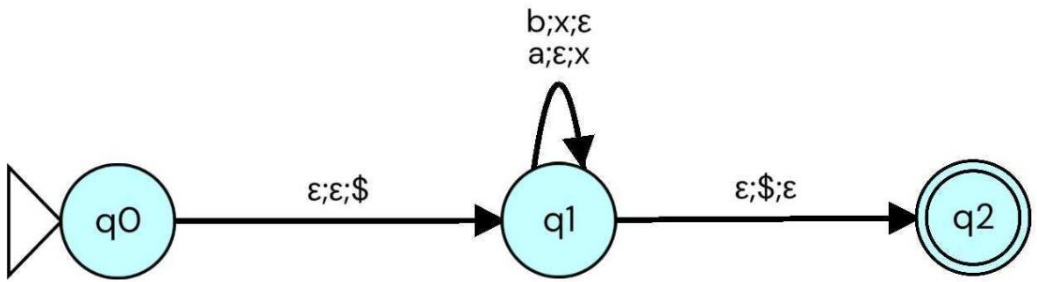
Examples for Creati
Pushdown Automat

- Let's Construct a PDA that accepts a
- L = {$a^n b^n$ }

This form is relevant when the order of alphabet is not important



Simulation: Pending



aabb
Accepted

aab
Rejected

abab
Accepted

+

FSAM

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- L = {w | n(a)=n(b)}

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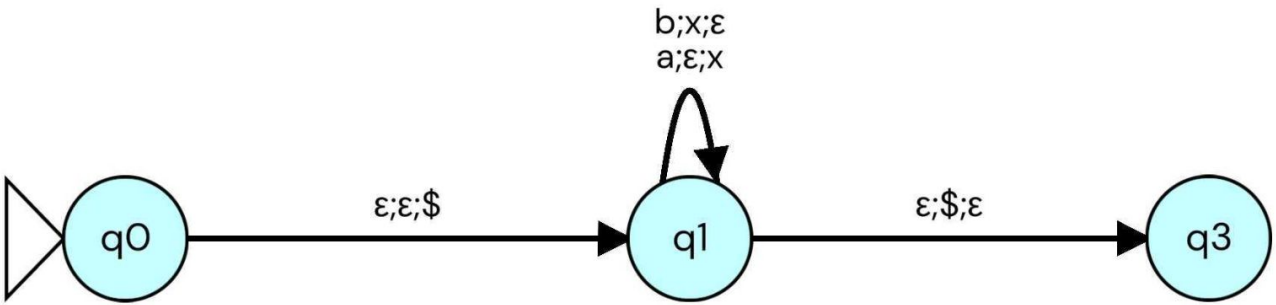
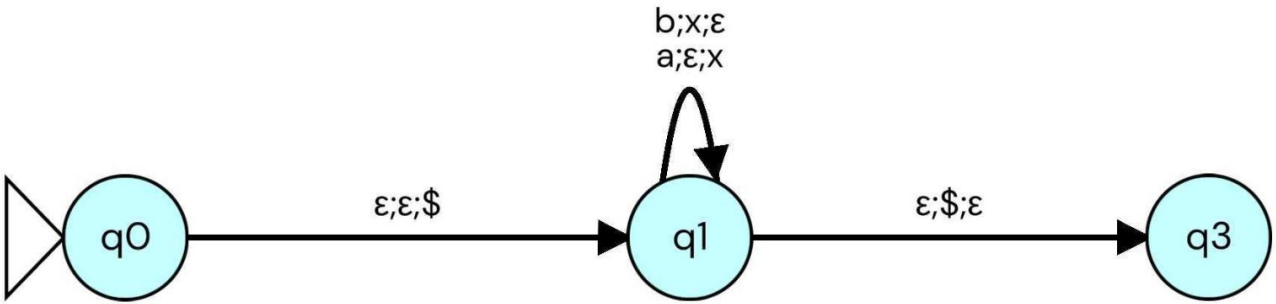


Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- L = {w | n(a)=n(b)}
  - **ba ?**



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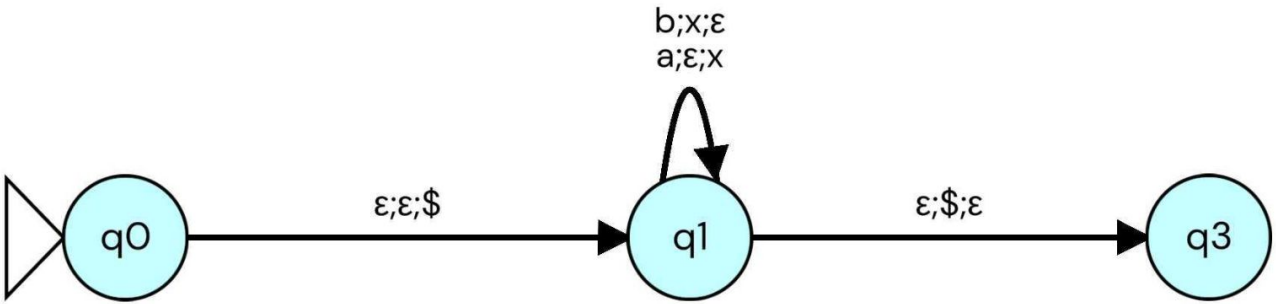
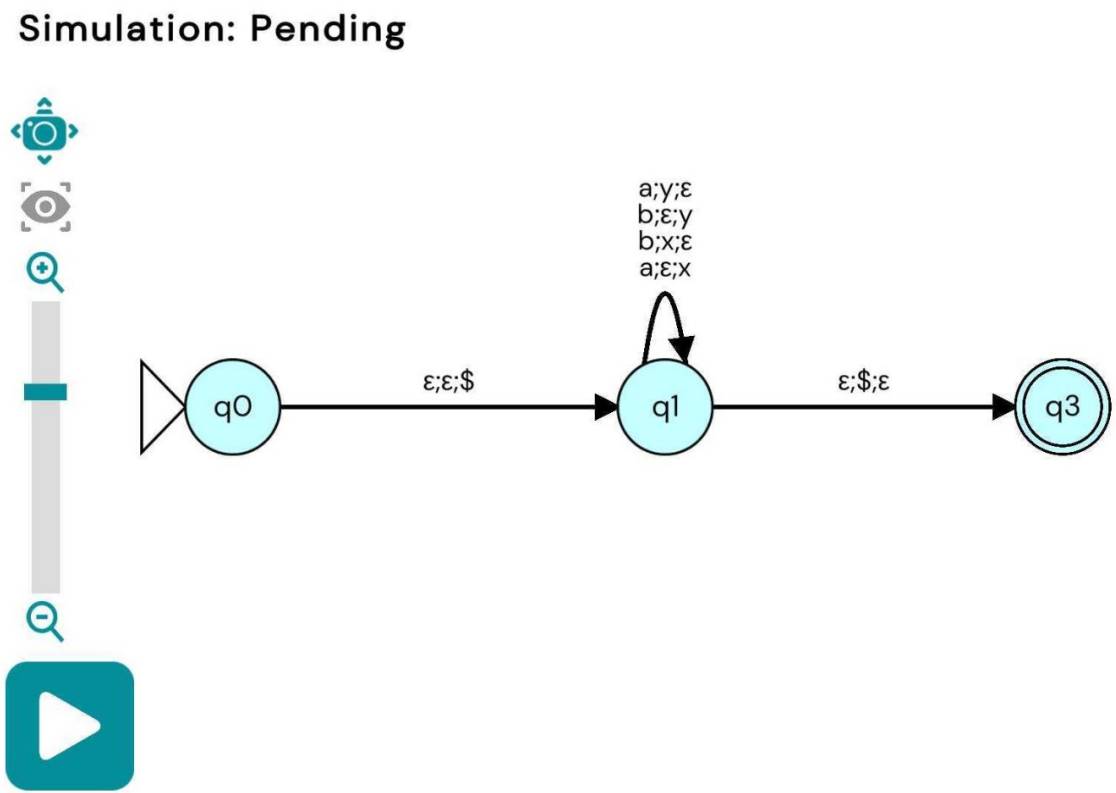
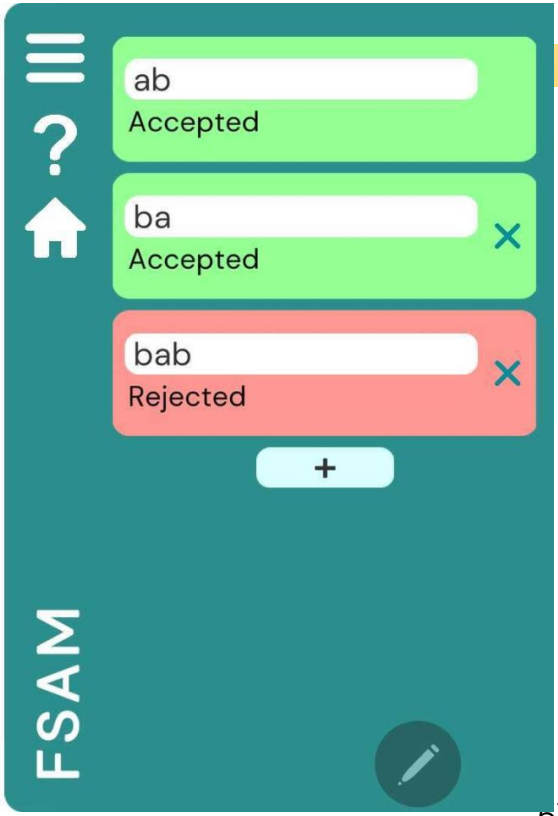


This should be accepting state ?

Examples for Creating Pushdown Automata





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- Let's Construct a PDA that accepts all strings from the language
- L = {w such that $w = w^R$ and w has an even length}

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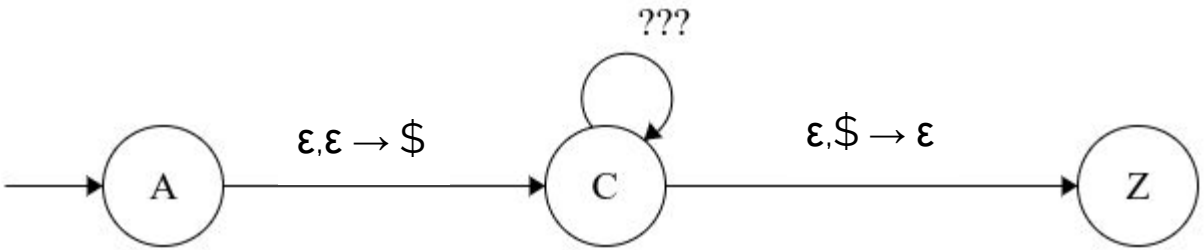

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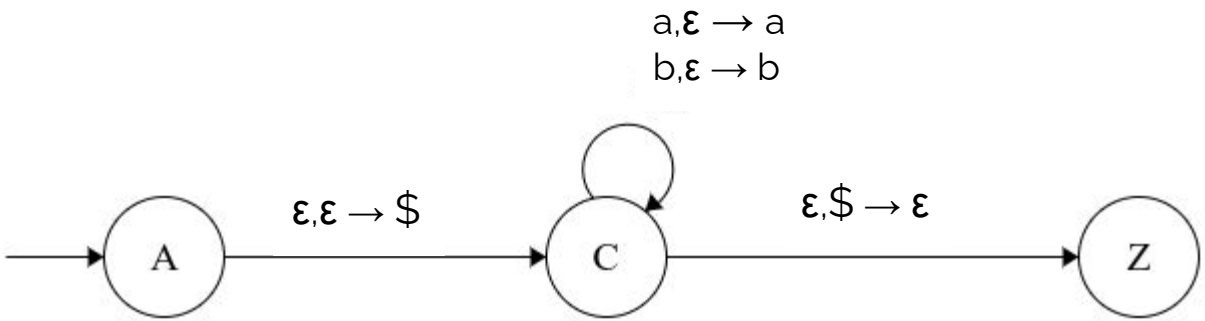

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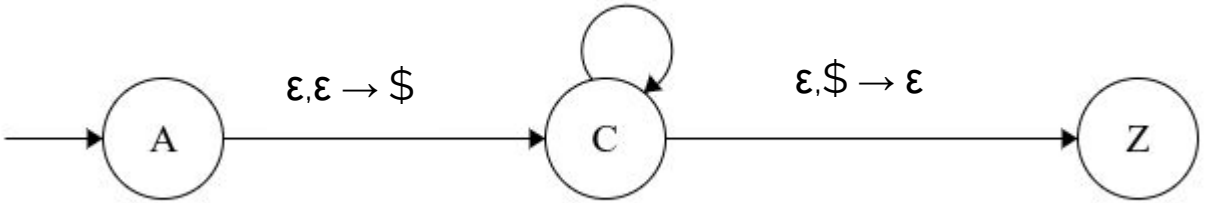
a,ε → a
b,ε → b
a,a → ε
b,b → ε



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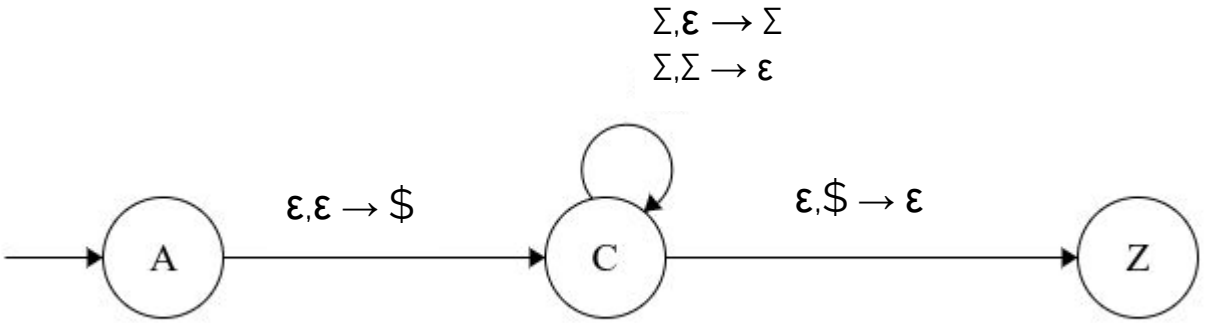

Examples for Creating Pushdown Automata

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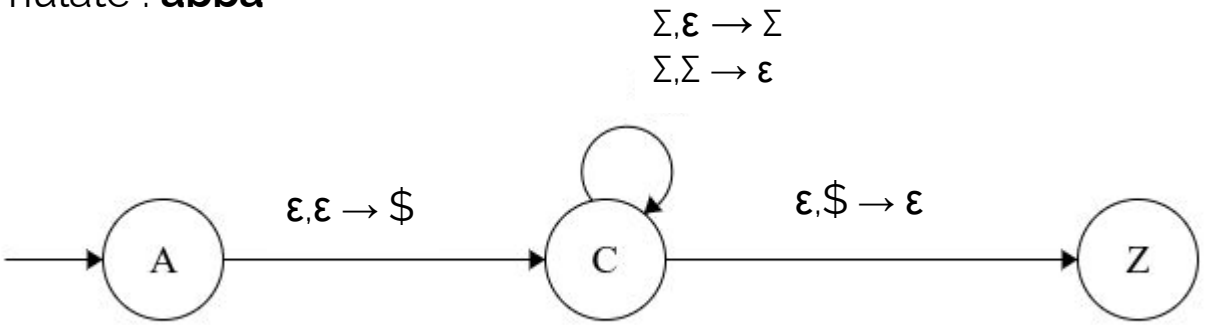
- Let's simulate : **abba**



Examples for Creating Pushdown Automata

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$\Sigma,\varepsilon \rightarrow \Sigma$
$\Sigma,\Sigma \rightarrow \varepsilon$

What about aabb ?

Examples for Creating Pushdown Automata

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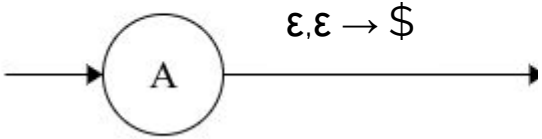

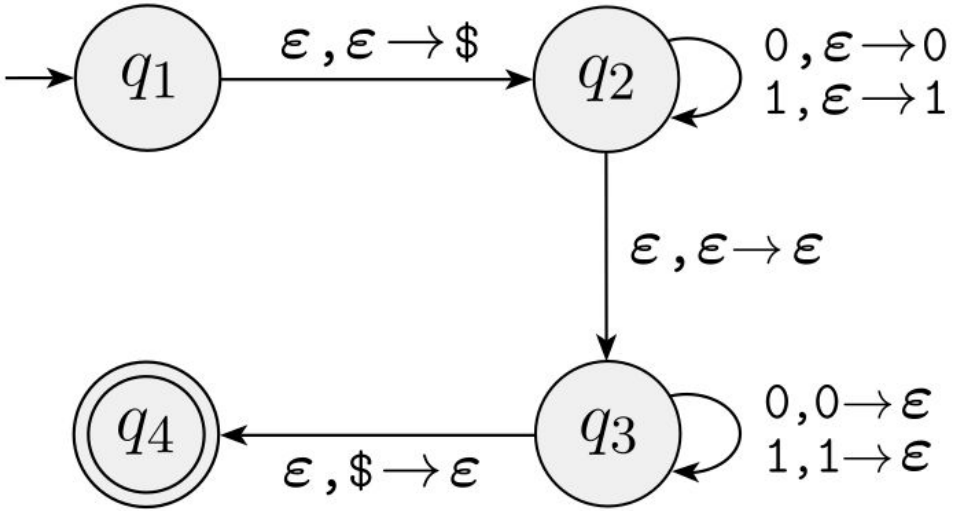

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- L = {w such that $w = w^R$ and w has an odd length}
  - aba
  - aabaa
  - baabaab

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- L = {w such that $w = w^R$ and w has an odd length}
  - aba
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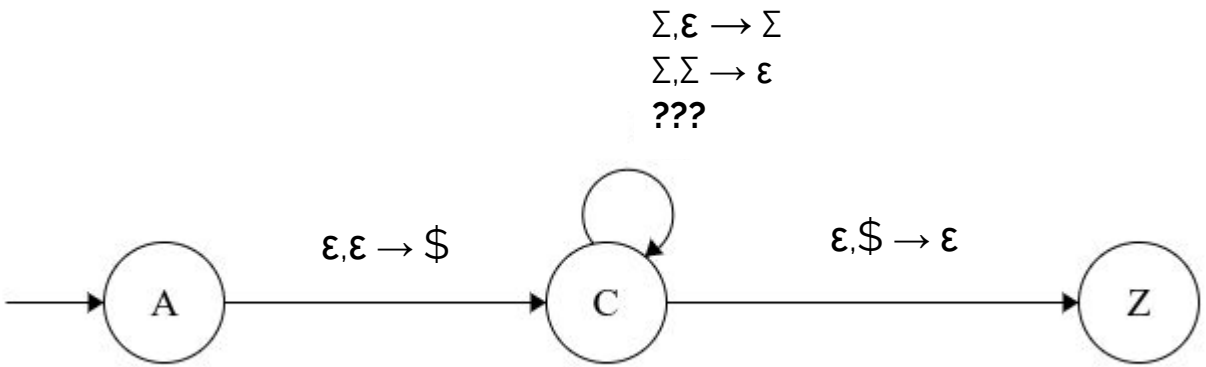

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- L = {w such that $w = w^R$ } { Odd + Even }

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- $L = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i = j \text{ or } i = k\}$

Examples for Creating Pushdown Automata

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Algorithm:

1. While next input character is a do push a

2. Nondeterministically, guess whether a's =b's or a's =c's

Case 1 : a's=b's

- While next input is b do pop a

- While next input character is c do nothing

Case 2 : a's=c's

- While next input is b do nothing

- While next input character is c do pop a

Examples for Creating Pushdown Automata

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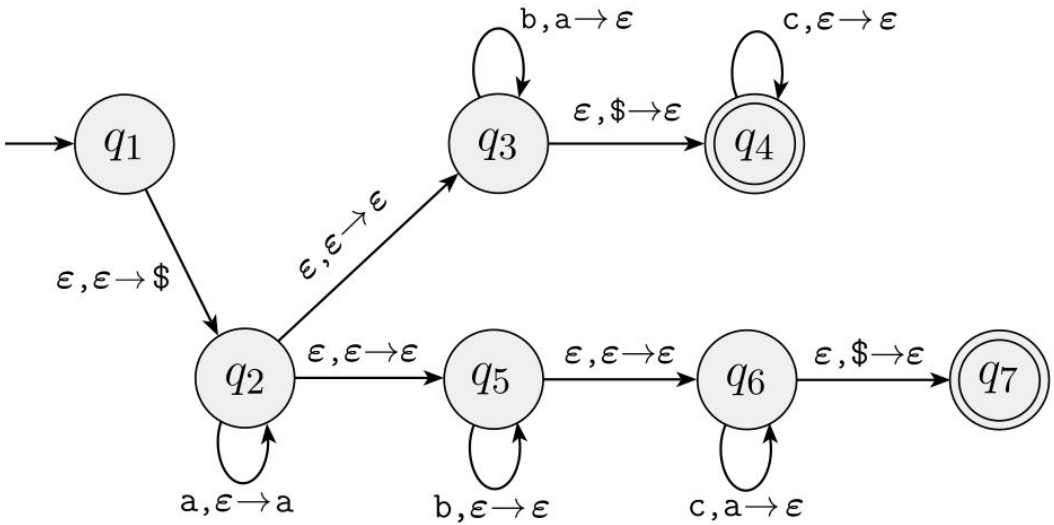


Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- L = {w | n(a)=n(b)+2}

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- $L = \{ww \mid w \in \{0,1\}^*\}$

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- $L = \{a^n b^m c^m d^n \mid w \in \{a,b,c,d\}^* , n,m>\}$

Examples for Creating Pushdown Automata

- Let's Construct a PDA that accepts all strings from the language
- Mathematical Expressions : ((1+2)*3)-4

Formalism for Pushdown Automata

- A pushdown automaton is a 6-tuple (Q, Σ, Γ, δ, q_0 , F), where Q, Σ, Γ, and F are all finite sets such that:
  1. Q is the set of states,
  2. Σ is the input alphabet,
  3. Γ (Gamma) is the stack alphabet,
  4. δ (Delta) : Q × Σε × Γε →P(Q × Γε) is the transition function,
  5. q_0 ∈ Q is the start state
  6. F ⊆ Q is the set of accept states.

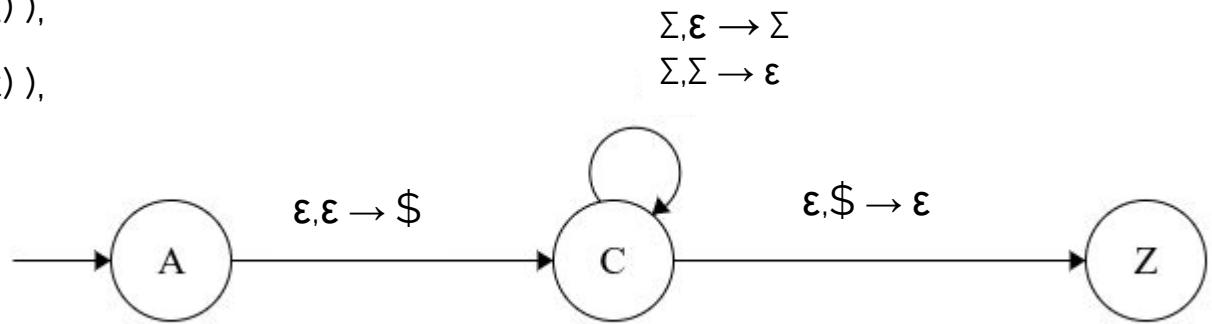
Formalism for Pushdown Automata

- Examples for Transitions :
  - $\delta : Q \times \Sigma\varepsilon \times \Gamma\varepsilon \rightarrow P(Q \times \Gamma\varepsilon)$ is the transition function
  - Δ = {

    ((A,ε,ε) , (C, $)),

    ((C,a,ε) , (C, a)),

    ((C,a,a) , (C, ε)),



Formalism for Pushdown Automata

- Examples for Transitions :
  - What's the language for these transitions:
  - Δ = {
    ((s, a, ε), (s, a))
    ((s, b, ε), (s, b))
    ((s, ε, ε), (f, ε))
    ((f, a, a), (f, ε))
    ((f, b, b), (f, ε))

    }

Formalism for Pushdown Automata

- Examples for Transitions :
  - $\delta : Q \times \Sigma\varepsilon \times \Gamma\varepsilon \rightarrow P(Q \times \Gamma\varepsilon)$ is the transition function
  - Δ = {

    ((A,**ε**,**ε**) , (C, $)),

    ((C,a,**ε**) , (C, a)),

    ((C,a,a) , (C, **aa**)),

Deterministic PDA

- For each state in the PDA, and for any combination of a **current input symbol** and a **current stack symbol**, there is at most one transition defined
- In other words, there is **precisely at most** one legal sequence of transitions that can be followed for any input
- What about the e-transitions ?

Taken from the slides of Keith Schwarz at Stanford

Deterministic PDA

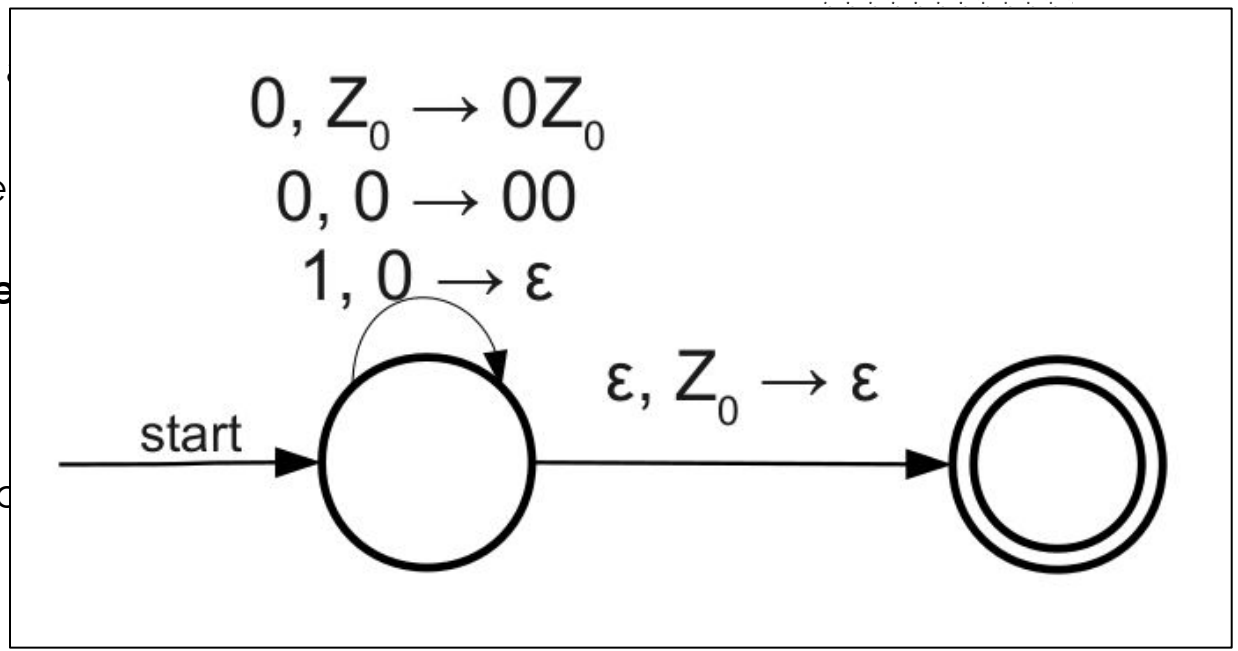
- For each state in the PDA,

  current stack symbol, the

- In other words, there is **pre**

  followed for any input

- What about the e-transitio



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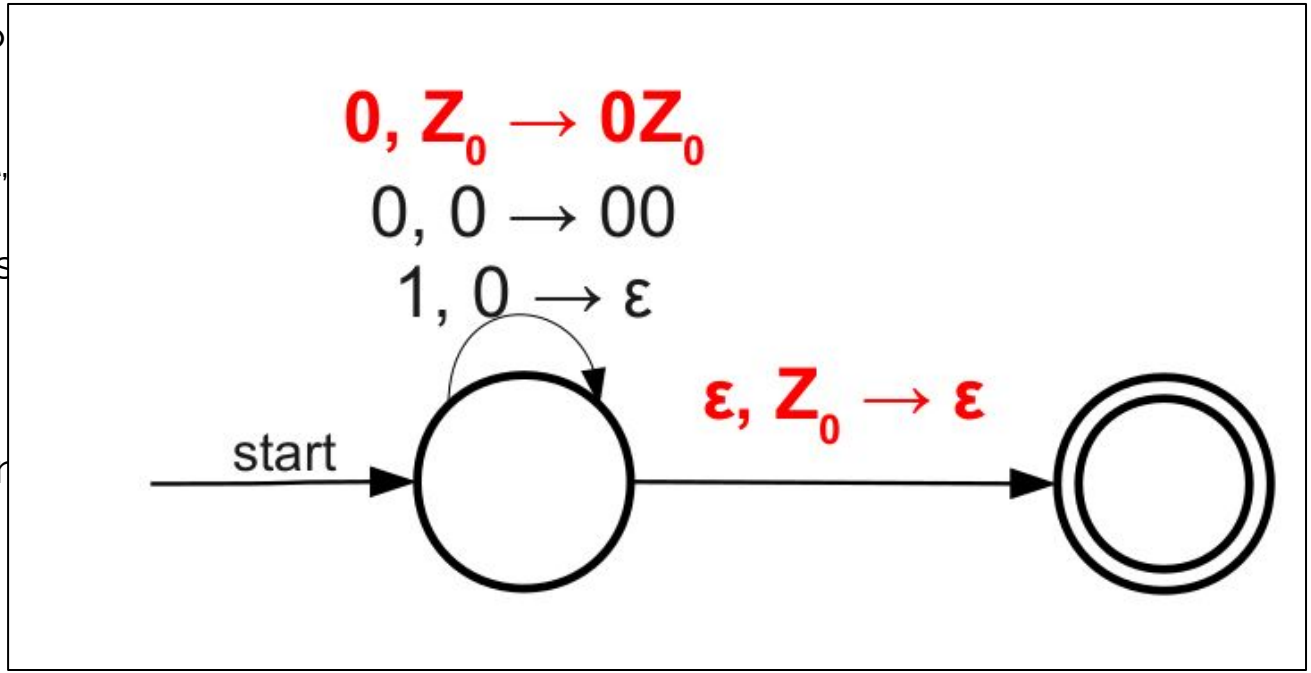

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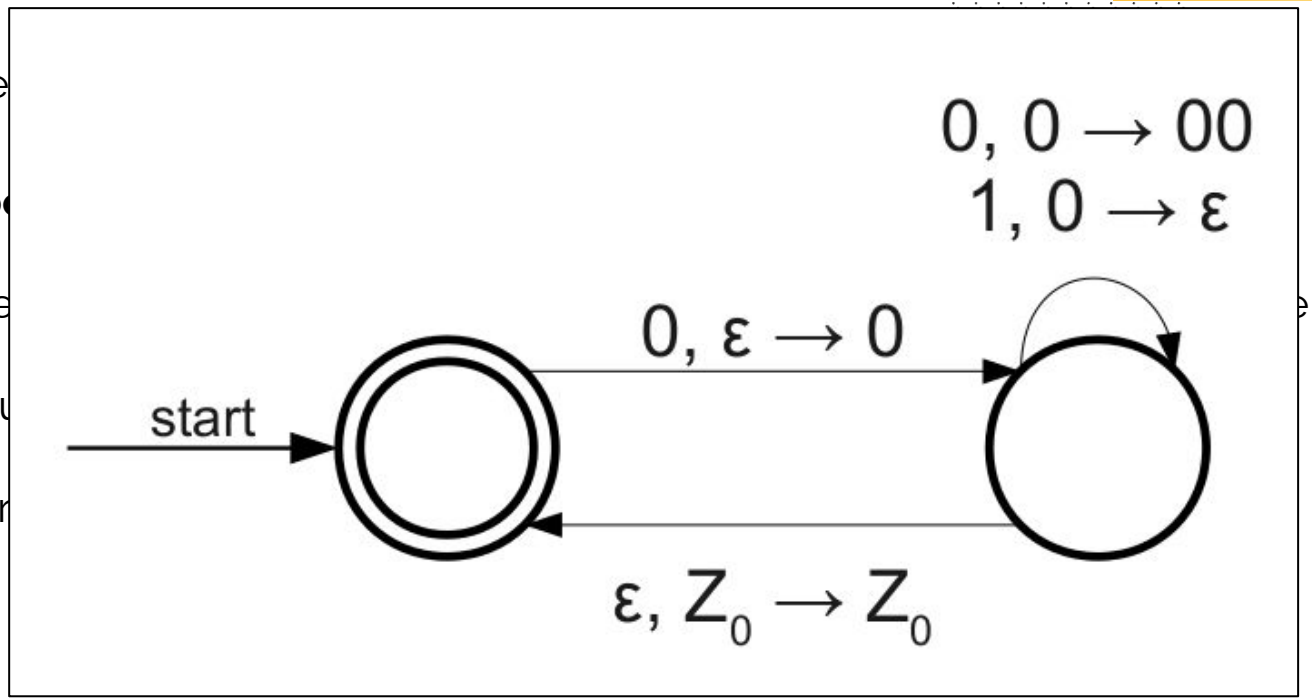

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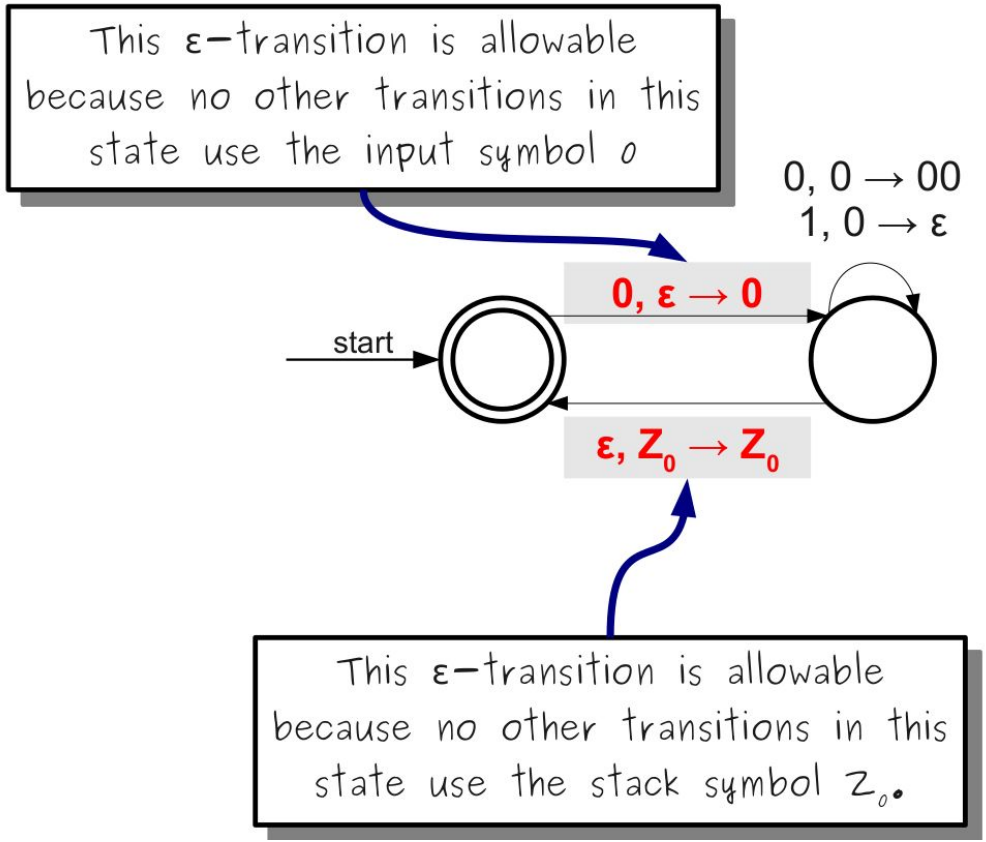

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Deterministic PDA

- If we can find a DPDA for a CFL, then we can recognize strings in that language efficiently.
- Can we guarantee that we can always find a DPDA for a CFL?

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Deterministic PDA

- If we can find a DPDA for a CFL, then we can recognize strings in that language efficiently.
- Can we guarantee that we can always find a DPDA for a CFL?
  - As DFA and NFA are equivalent and each NFA has its DFA equivalent
  - Do PDA and DPDA have the same power ?
    - Does any CFL represented by a DPA, has an DPDA equivalent ?

Deterministic PDA

- Simple example: **The language of palindromes.**
- Design the algorithm for the DPDA
- How do you know when you've read half the string?
    - It is deterministic, the machine does not have the power for guessing or branching …

Equivalence

- A language is context free if and only if some pushdown automaton recognizes it.

    1. **If a language is context free, then some pushdown automaton recognizes it.**
    2. If a pushdown automaton recognizes some language, then it is context free.

Equivalence : CFG → PDA

- Simple Question : How to find whether a word can be generated by a CFG ?

Equivalence : CFG → PDA

- Simple Question : How to find whether a word can be generated by a CFG ?
  - We start with the start variable S
  - Find a suitable production rule for S and replace.
  - Recursively, For each variable, find a suitable production rule
  - If a terminal is reached and matching the actual char on the string to assess, move on to the next part.

Equivalence : CFG → PDA

- Simple Question : How to find whether a word can be generated by a CFG ?
  - We
  - Fi
  - Re
  - If ss,

    move on to the next part.

How can we automate such process ?

How can we create a machine for this ?

How to translate a CFG into a PDA ?

Equivalence : CFG → PDA

- Given the following grammar :
  - S → aS | **ε**
- **For simplification, we derive the following word:**
  - **aa**

Equivalence : CFG → PDA

- Simple Idea :
  - **Push Variables into the stack**
  - **Replace Top Variable by its Production rules into the stack**

Equivalence : CFG → PDA

- Given the following grammar :
  - S → aS | **ε**
    - **Initialize the stack with the marker symbol $**
    - **Place the start variable at the top of the Stack**

S
$

Equivalence : CFG → PDA

- Given the following grammar :
  - S → **aS** | **ε**
    - **We replace the variable with its production rule**

      S → **aS**

      Within the stack

a
S
$

aa

Equivalence : CFG → PDA

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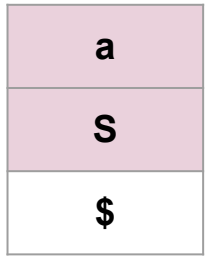
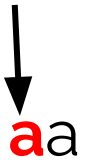
      S → **aS**

      Within the stack





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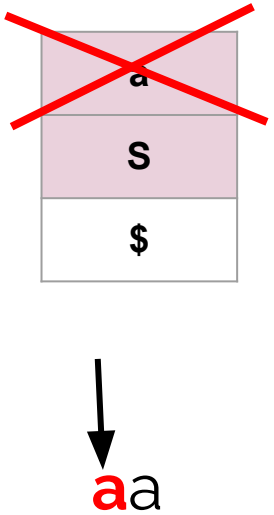
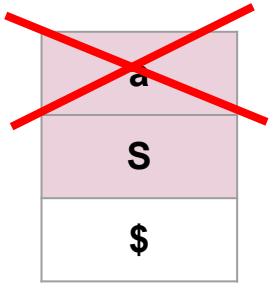
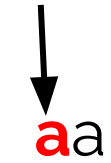


Equivalence : CFG → PDA

- Given the following grammar :
  - S → **aS** | **ε**
    - **Terminal Symbol “a” must be POPPED from the stack**
    - **Equivalent in PDA :**

      The transition should be made as : a, a → ε



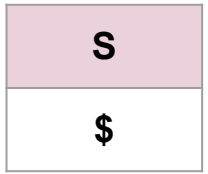


Equivalence : CFG → PDA

- Given the following grammar :
  - S → **aS** | **ε**
    - **We still have one letter**

      AND

    - **For Variable S on top of Stack ? how can we do its transition on a PDA ?**



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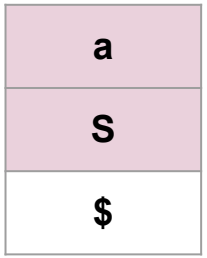

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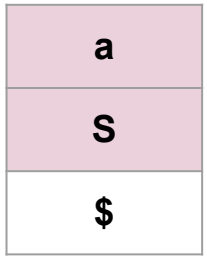
      - **ε , S → aS**



a

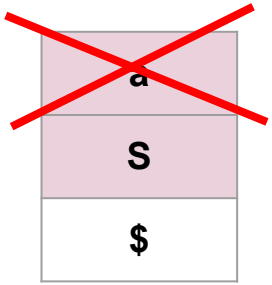
We push two letters into the Stack

Equivalence : CFG → PDA

- Given the following grammar :
  - S → **aS** | **ε**
    - **We still have one letter**

      AND

    - **For Variable S on top of Stack ? how can we do its transition on a PDA ?**
      - **ε , S → aS**



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- Given the following grammar :
  - S → **aS** | **ε**
    - **We still no letter , we have matched all letters from**

      From the given string (aa)

      AND
    - **For Variable S on top of Stack ? how can we do its transition on a PDA ?**

S
$

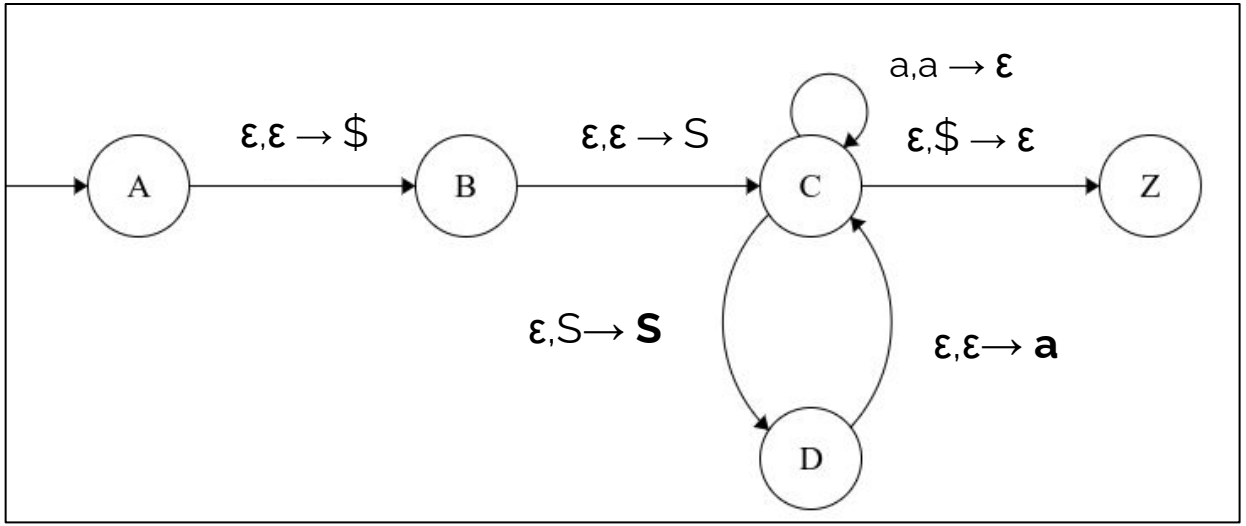
Equivalence : CFG → PDA

- Given the following grammar :
  - S → **aS** | **ε**
    - **We choose the substitution rule:**
      - S → **ε**
    - **For Variable S on top of Stack ? how can we do its transition on a PDA ?**
      - **ε , S → ε**

S
$

Equivalence : CFG → PDA

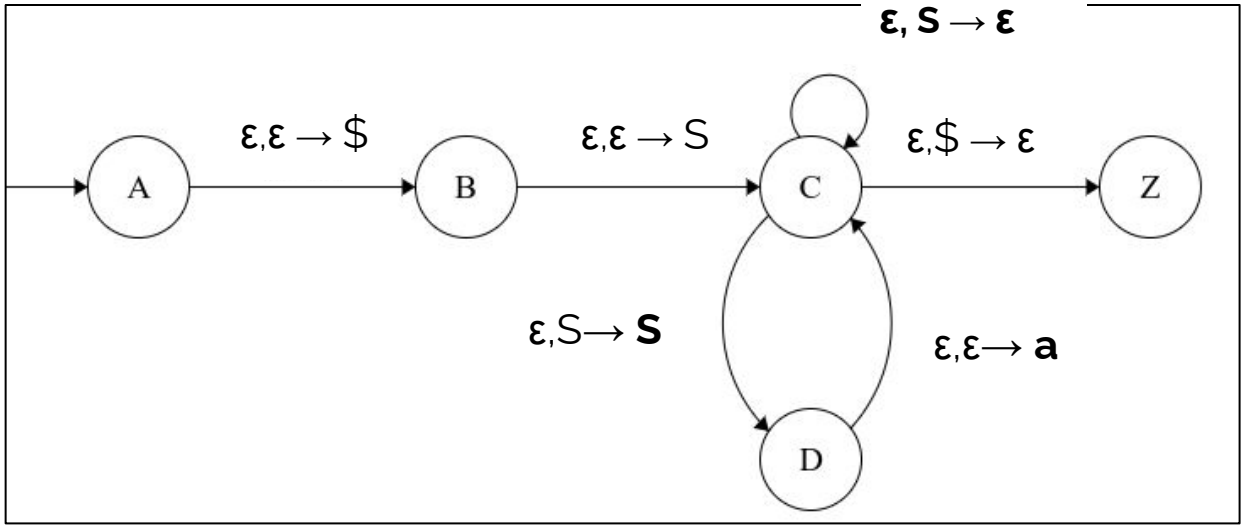
- Given the following grammar :
  - S → **aS** | **ε**



a
S
$

Equivalence : CFG → PDA

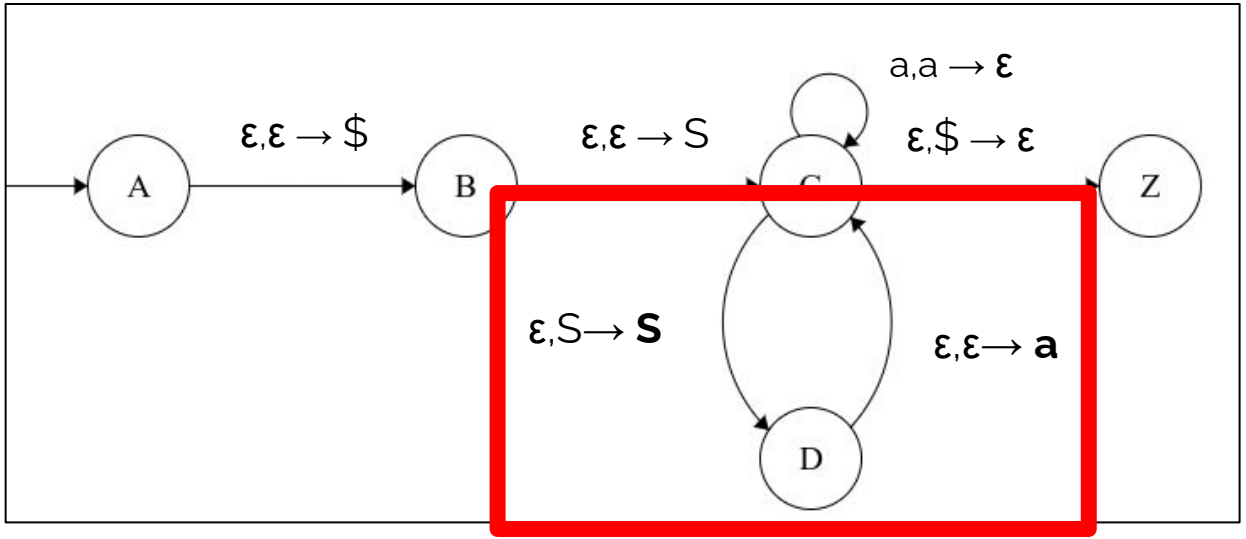
- Given the following grammar :
  - S → **aS** | **ε**



a
S
$

Equivalence : CFG → PDA

- Given the following grammar :
  - S → **aS** | **ε**



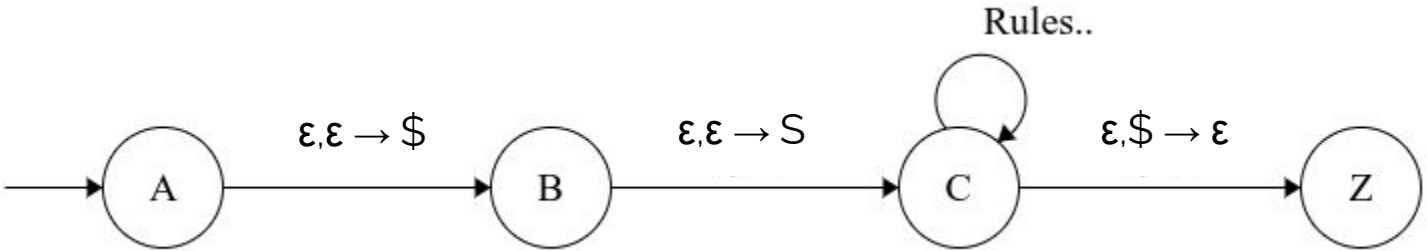
a
S
$

Equivalence : CFG → PDA

- Given the following grammar :
  - $S \rightarrow X \mid \varepsilon$
  - $X \rightarrow aXb \mid \varepsilon$

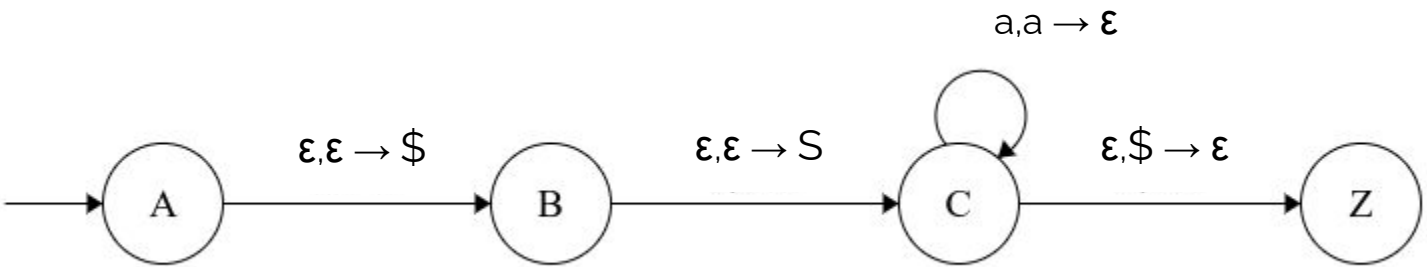
Equivalence : CFG → PDA

- Given the following grammar :
  - S → X | ε
  - X → aXb | ε



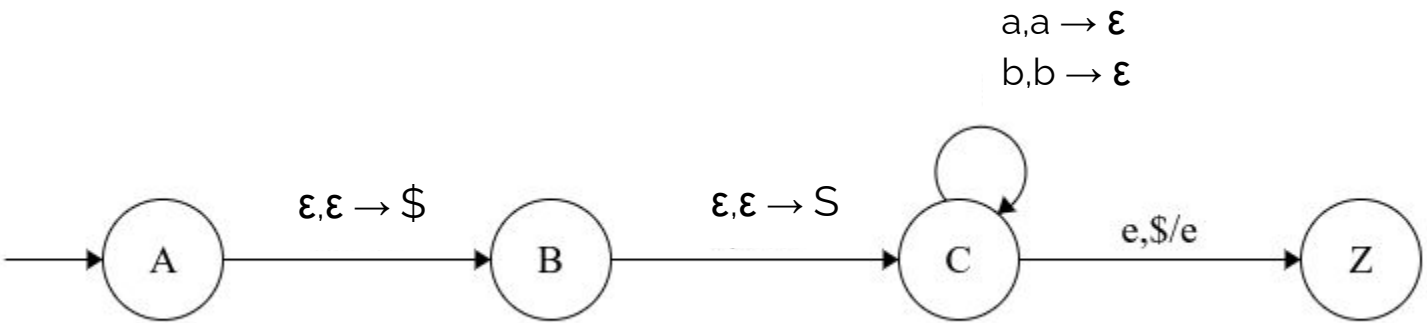
Equivalence : CFG → PDA

- Given the following grammar :
  - S → X | ε
  - X → aXb | ε



Equivalence : CFG → PDA

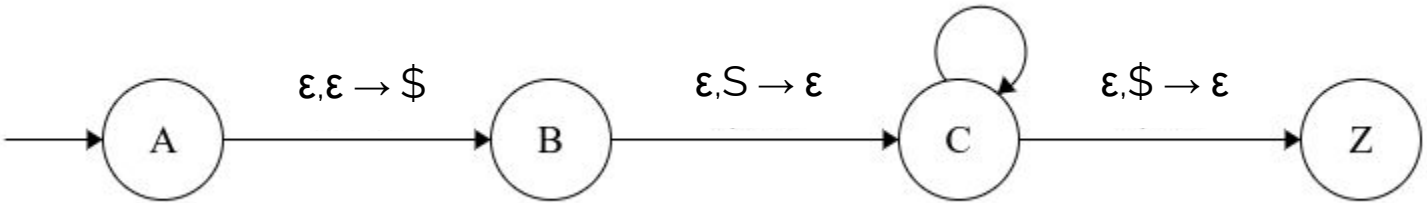
- Given the following grammar :
  - S → X | ε
  - X → aXb | ε



Equivalence : CFG → PDA

- Given the following grammar :
  - S → X | ε
  - X → aXb | ε

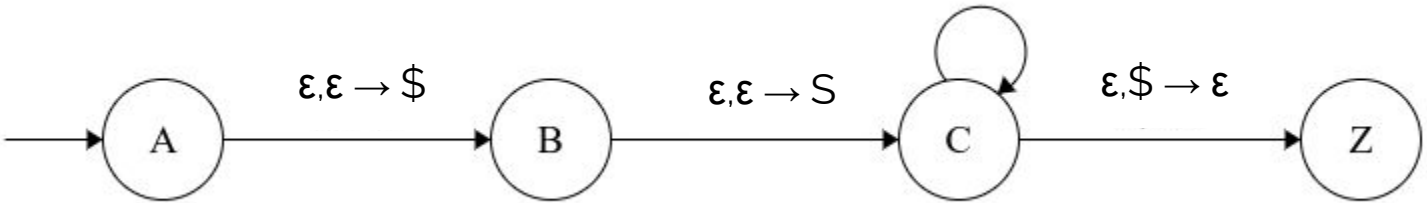
a,a → ε
b,b → ε
ε, S → X
ε, X → aXb



Equivalence : CFG → PDA

- Given the following grammar :
  - S → X | **ε**
  - X → aXb | **ε**

a,a → **ε**
b,b → **ε**
ε, S → X
ε, X → aXb
ε,S → ε
ε,X → ε

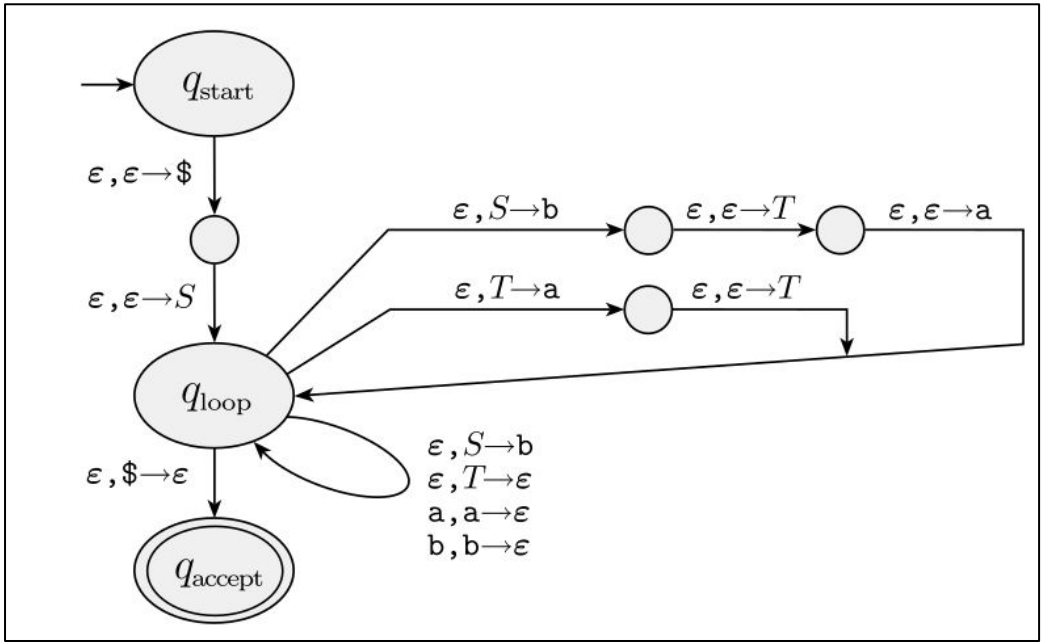


Equivalence : CFG → PDA

- Given the following grammar :
  - S → a T b | b
  - T → Ta | ε

Equivalence : CFG → PDA

- Given the following grammar :
  - S → a T b | b
  - T → Ta | ε



Equivalence : PDA → CFG

- A language is context free if and only if some pushdown automaton recognizes it.

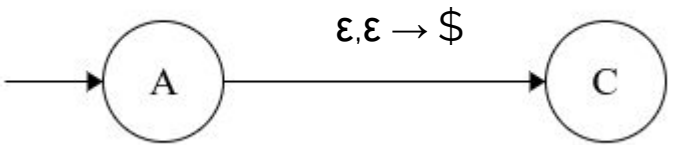
  1. If a language is context free, then some pushdown automaton recognizes it.
  2. **If a pushdown automaton recognizes some language, then it is context free.**

     (Steps will not be covered in this course)

Equivalence : PDA → CFG

- The Algorithm:

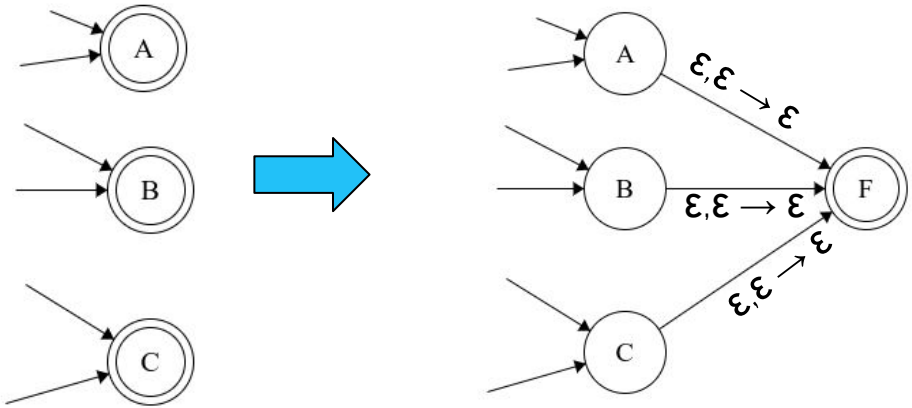
1. Simplify the PDA:
   - *Create a new Start State and initialize the stack with **$***



Equivalence : PDA → CFG

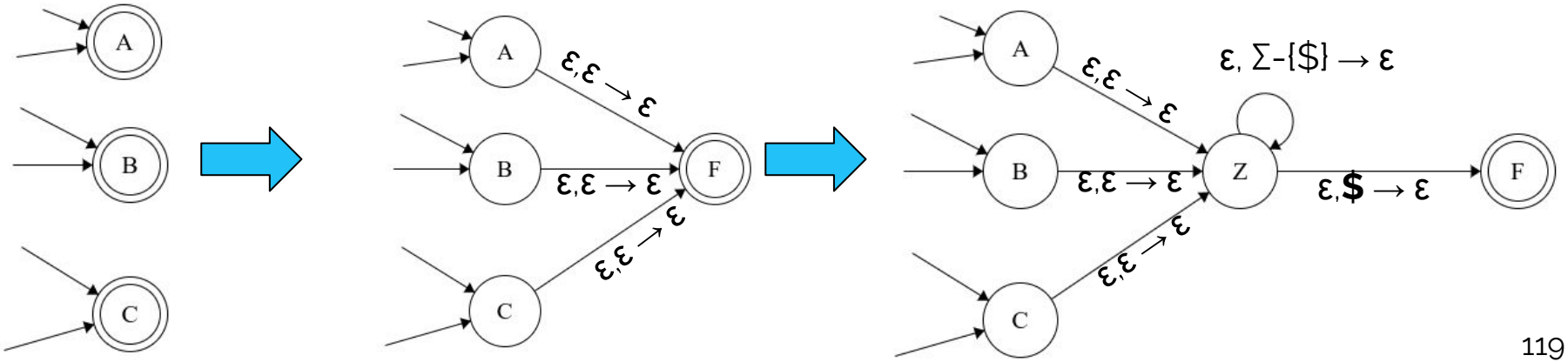
- The Algorithm:

1. Simplify the PDA:
   - *Should have only one accept state newly created*



Equivalence : PDA → CFG

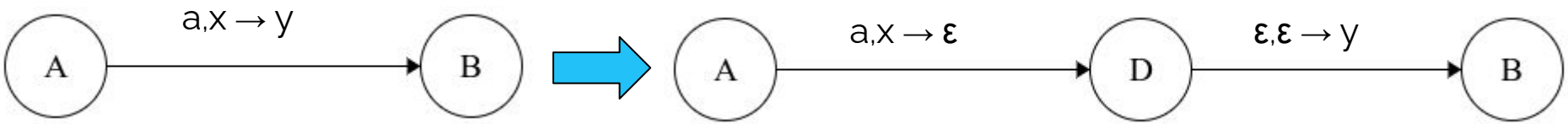
- The Algorithm:
  1. Simplify the PDA:
     - *The stack needs to be **emptied** just after passing the newly created final state*



Equivalence : PDA → CFG

- The Algorithm:

1. Simplify the PDA:
   - *Transform all transitions so that each transition would do at a time either :*
     - *push a symbol or*
     - *Pop a symbol*



Equivalence : PDA → CFG

- The Algorithm:

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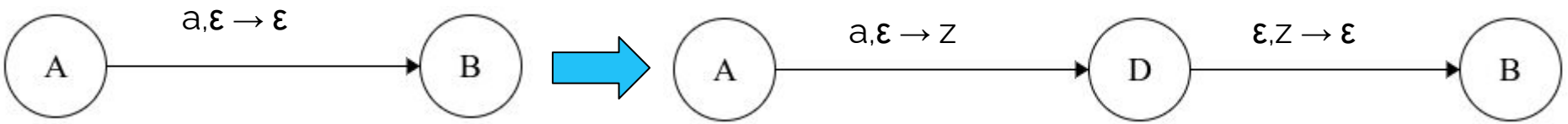


Equivalence : PDA → CFG

- The Algorithm:

2. Construct the Context Free Grammar

   - *For each **reachable/traversable** pair of states (A, B), create a variable (non-terminal symbol) V_{AB}*
   - *The start variable is V_{SF} such that S is the start state and F is the final state*
   - *Create Production Rules based on the following cases….*

Equivalence : PDA → CFG

- The Algorithm:

  2. Construct the Context Free Grammar

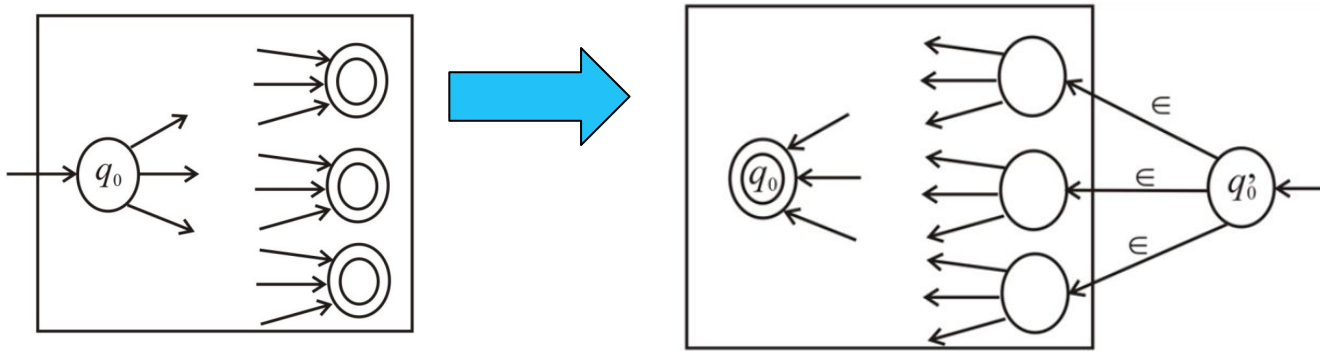
     - *Create Production Rules based on the following cases*

       - *if* $\delta(p, a, \varepsilon) \rightarrow (r, u)$ *and* $\delta(s, b, u) \rightarrow (q, \varepsilon)$*, add the following rule : G:* $A_{pq} \rightarrow aA_{rs}b$
       - *For each p, q, r* $\in Q$*, add the following rule to G:* $A_{pq} \rightarrow A_{pr}A_{rq}$
       - *For each p* $\in Q$*, add the following rule to G:* $A_{pp} \rightarrow \varepsilon$

TD5 - Solutions

- **Exercise 1**

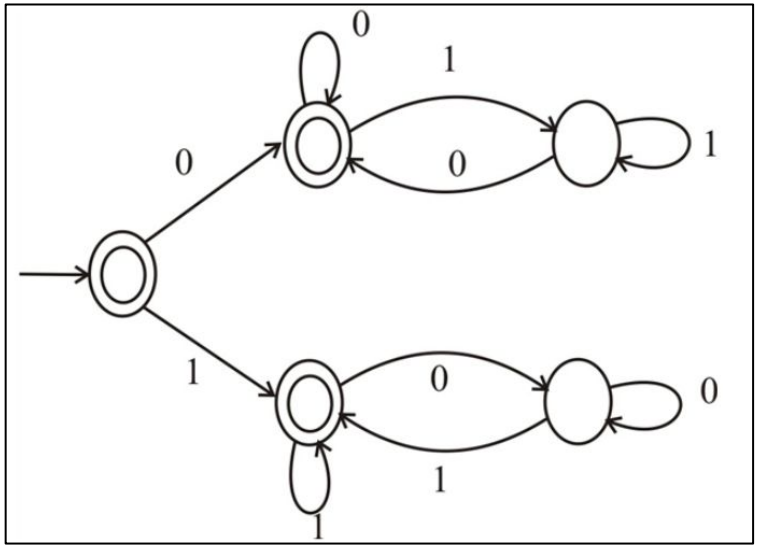
1. For any string $w = w_1 w_2 \cdots w_n$, the reverse of w, written w^R, is the string w in reverse order, $w_n \cdots w_2 w_1$. For any language A, let $A^R = \{w^R \mid w \in A\}$. Show that if A is regular, so is A^R.
   - Done by transforming the NFA for this language to as follows:
     - Invert the direction of all transitions
     - Create a new start state q0' and link them to the accepting states
     - Invert all accepting states into non-accepting states
     - Set the original start state as an accepting state.



TD5 - Solutions

- **Exercise 1**

2. Let Σ = {0,1} and let D = {w| w contains an equal number of occurrences of the substrings 01 and 10}. Thus 101 ∈ D because 101 contains a single 01 and a single 10, but 1010 not in D because 1010 contains two 10s and one 01. Show that D is a regular language.



TD5 - Solutions

- **Exercise 2**
  - Prove that the following languages are non-regular
    1. $A_1 = \{0^n 1^n 2^n \mid n \geq 0\}$
    2. $A_2 = \{www \mid w \in \{a, b\}^*\}$
    3. $A_3 = \{a2n \mid n \geq 0\}$ (Here, a2n means a string of 2^n a's.)
    4. $\{w \mid w \in \{0,1\}^*$ is not a palindrome$\}$ is not a regular

TD5 - Solutions

- **Exercise 2**
  - Prove that the following languages are non-regular

1. $A_1 = \{0^n 1^n 2^n \mid n \geq 0\}$
   a. We consider that the language A_1 is regular
   b. Therefore, there must be a pumping mechanism.
   c. We assume the pumping constant is **P**
   d. We take the string $0^P1^P2^P$ which is in the language,
   e. There are infinitely many words, and words with larger sizes that can be generated even from this word : $0^P1^P2^P \rightarrow (0^{2P}1^{2P}2^{2P}\ldots\ldots)$
   f. The word can be written in the form s=xyz such that |xy|<=P and |y|=k>0
      i. xy must be in the part of 0^P
      ii. Y must be only in the zero part.
      iii. If we pump Y, the new word will be in the form $0^P0^K1^P2^P = 0^{P+K}1^P2^P$
      We will **always have words not** in the language, as zeros will be more than 1 and 2.
   g. Therefore, we cannot pump more words from S to have new words in the language.
   h. ⇒ No pumping mechanism.
   i. The language is not regular

TD5 - Solutions

- **Exercise 2**
  - Prove that the following languages are non-regular

1. $A_1 = \{0^n 1^n 2^n \mid n \geq 0\}$
2. $A_2 = \{www \mid w \in \{a, b\}^*\}$ ⇒ $\mathbf{a^p b a^p b a^p b}$
3. $A_3 = \{a^{2^n} \mid n \geq 0\}$ (Here, a2n means a string of 2^n a's.) ⇒ a^2^P
4. {w| w ∈ {0,1}* is not a palindrome} is not a regular ⇒ $0^p 1 0^p$

TD5 - Solutions

- **Exercise 2**
  - Prove that the following languages are non-regular
    1. A_3 = {a^2^n | n ≥ 0} (Here, a2n means a string of 2^n a's.) ⇒ a^2^P
       a. a^(2^p)
          i. Example for simplification only not to fix P at a given number :
             1. P=4 ⇒ aaaa …aaa (2x2x2x2 ⇒ 16 times)
             2. Next word is P+1=5 ⇒ aaa …aaa (2x2x2x2x2 ⇒ 32 times)
             3. ..
          ii. The word can be written as s=xyz such that |xy| <=P and |y|=K>0
          iii. Regardless of the value of y if we would like to generate the next word by pumping.
             1. We need to have the next word which must be in the language,
             2. $|xyz|=2^P$
             3. The next word is of course : xy^2z such that $\mathbf{|xy^2z|=2^{P+1}}$
             4. But as |xy| <=P even x is empty, and y is all the **a**s (by considering even |Y|=P at max)
                a. $|xy^2z| <= 2^p+p$ but
                b. $2^p+p < 2^{p+1}$ which is true always by induction
                c. Therefore, the new word will never be in the language.

TD5 - Solutions

- **Exercise 2**
  - Prove that the following languages are non-regular
    1. A_3 = {a^2^n | n ≥ 0} (Here, a2n means a string of 2^n a's.) ⇒ a^2^P
       a. **$2^p+p < 2^{p+1}$**
          - We assume that $2^p+p < 2^{p+1}$ is true
          - We multiply by both sides by 2
          - $2(2^p+p) < 2^*2^{p+1}$
          - $2^{p+1}+2p < 2^{(p+1)+1}$
          - For P >= 1, it is always , p+1<p+p<2p,
            - Therefore, we can replace 2p with a lesser number inside the smaller side of the inequality.
          - $2^{p+1}+ p+1 < 2^{p+1}+ 2p < 2^{(p+1)+1}$
          - $2^{p+1}+ p+1 < 2^{(p+1)+1}$
          - Therefore, always true by induction

TD5 - Solutions

- **Exercise 3**
  - Let B = {a^k | k is a multiple of n}. Show that for each n ≥ 1, the language B_n is regular.
    - B_1 = a, we can write regular expressions = a
    - B_2 = (aa)*
    - B_3 = (aaa)*
    - ..
    - B_n= (aa…aa)* (a is repeated n times)
    - B is the union of B_1, B_2,...B_n is regular as the union of regular languages is regular.

TD5 - Solutions

- **Exercise 3**
  - For languages A and B, let the perfect shuffle of A and B be the language $\{w \mid w = a_1 b_1 \cdots a_k b_k$, where $a_1 \cdots a_k \in A$ and $b_1 \cdots b_k \in B$, each a_i , $b_i \in \Sigma\}$.
    Show that the class of regular languages is closed under a perfect shuffle.
    Example : abc ∈ A, 123 ∈ B , by perfectly shuffling → a1b2c3
  - The new language S will be constructed from A and B by taking words of the same size and taking a letter from each word in an alternating fashion.
    - If the states of the DFA for A is X={x0, x1, x2,x3…xn}
    - If the states of the DFA for B is Y={y0, y1, y2,y3…yn}
    - The DFA Machine can be constructed for the language S with the following states
      - X * Y * {A,B }
      - Examples
        - (x0,y1,A) (I am now at machine A, at state x0 whilst i was at state of y1 of B)
      - Start state would be: (x0,y0, A)
      - Accepting States would be:
        - $(x_{accept}, y_{accept}, A)$
      - Transitions would be:
        - δ((xn, yn, A) , a) → (δ(xn, a) , yn , **B**)
        - δ((xn, yn, B) , a) → (xn, δ(yn, a) , **A**)

TD6 - Solutions

In each case below, say what language is generated by the context-free grammar:

1. S → aS | bS | ε **{a,b}***

2. S → SS | bS | a **{a,b}*a**

3. S → SaS | b **starts with b and ends with b + a and b are alternating**

4. S → SaS | b | ε **does not contain bb**

5. S → T T **contains exactly two bs**
   T → aT | T a | b

6. S → aSa | bSb | aAb | bAa **not palindromes**
   A → aAa | bAb | a | b |ε | S

7. S → aT | bT | ε **Even number of letters**
   T → aS | bS

8. S → aT | bT **odd number of letters**
   T → aS | bS | ε

TD6 - Solutions

Give the context-free grammars that generate the following languages. Alphabet Σ is {0,1}.

1. {w| w contains at least three 1s}
   S→P1P1P1P
   P → 0P | 1P | e

2. {w| w starts and ends with the same symbol}
   S → 0 P 0 | 1 P 1 | 1 | 0
   P → 0P | 1 P | e

3. {w| the length of w is odd}
   S→ 0 | 1 | 00S | 10S | 10S | 11S
   Or
   S→ 0 | 1 | 0S0 | 0S1 | 1S0 | 1S1
   Or
   See previous exercise

4. {w| the length of w is odd and its middle symbol is a 0}
   S → 0 | 0S0 | 0S1 | 1S0 | 1S1

TD6 - Solutions

Give the context-free grammars that generate the following languages. Alphabet Σ is {0,1}.

1. {w| w = w^R , that is, w is a palindrome}

   S → 0S0 | 1S1 | 1 | 0 | ε

2. {w| w is not equal to w^R , that is, w is not a palindrome}

   S → 0S0 | 1S1 | 0A1 | 1A0
   A → 0A0 | 1A1 | 0 | 1 | ε | S

3. {number of 0 is the same as 1}

   S → ε | S0S1S | S1S0S
   OR
   S → 0S1 | 1S0 | SS | ε

4. All strings with more a's than b's

   S → S_1aS_1
   S_1 → $bS_1a|aS_1b|S_1S_1|aS_1|$ ε

   Test String : aabbaa :

   S–> S_1aS_1 → aS_1b aS_1 → aaS_1bb aS_1 → aaS_1bb aS_1→ aabb aS_1 → aabbaS_1 → aabbaa